

Recreational Marijuana Laws and Workplace Absence^{*}

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Abstract

A near majority of states have now passed recreational marijuana laws (RMLs), but their impact on labor market behavior remains unclear. This study provides new causal evidence on the relationship between RMLs and workplace absence. Using a generalized difference-in-differences framework to exploit the staggered rollout of RMLs, I first document that the onset of legal recreational sales is accompanied by a 50% increase in the prevalence of prior-month adult marijuana use in treated states relative to control states. Next, I show that retail market access leads to a 14% increase in the incidence of prior-week workplace absence. Supplemental analyses of health outcomes suggest that part of the effect may be driven by increases in self-reported mental health problems following the opening of recreational dispensaries. These findings suggest that marijuana legalization may affect labor market outcomes along margins not typically captured by employment status alone.

Keywords: Recreational Marijuana Laws, Workplace Absence, Employment, Health

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1 Introduction

The landscape of marijuana legalization in the United States has transformed considerably since 2012, with 23 states and the District of Columbia enacting recreational marijuana laws (RMLs) by September 2023. These laws permit adults aged 21 and older to possess, purchase, and, in many cases, cultivate marijuana for non-medical use, thereby expanding legal access to an estimated 158 million Americans as of 2022 (Dave et al., 2025). Public support for such laws has also grown, reaching 68% in 2020 (Gallup, 2020). Proponents of legalization emphasize potential economic benefits, including job creation, increased tax revenue, and the growth of a new legal industry, which advocates argue could support economic recovery efforts (J. Brown et al., 2023).

However, prominent health organizations such as the American Medical Association and the American Public Health Association have refrained from endorsing RMLs, citing unresolved concerns about public health consequences (American Public Health Association, 2020; Bailey, 2021). Critics contend that increased accessibility to recreational marijuana may contribute to workplace absenteeism (Yang et al., 2024). Frequent use has been associated with cognitive impairment (Hanson et al., 2010; Hoch et al., 2025) and motivational deficits (Volkow et al., 2016), both of which are linked to lower workforce participation and engagement. The CDC has called for more research on labor market impacts, including how cannabis use affects workers' abilities to perform their jobs, noting increased absenteeism as one possible impact (Howard et al., 2020).

Concerns have also been raised regarding the potential rise in cannabis use disorder and heavy marijuana consumption, which may further affect labor force participation and productivity (Anderson & Rees, 2023; J. Brown et al., 2023). Some hypothesize that disordered marijuana use could increase the risk of developing amotivational symptoms (Dave et al., 2025; Lawn et al., 2016; Volkow et al., 2016) and lethargy (Dave et al., 2025; Irons et al., 2014; Pesta et al., 2013), which may impact absenteeism. Furthermore, physical and psychological health risks associated with frequent marijuana use (National Academies of Sciences, Engineering, and Medicine et al., 2017) could lead to increased sick leave and reduced workplace productivity. If RMLs amplify these impacts,¹ the resulting absenteeism and declining labor force engagement could have significant implications for labor market performance and economic stability.

Despite the importance of understanding the labor market impacts of cannabis legalization, prior studies have mostly focused on extensive margin outcomes such as employment (Anderson & Rees, 2023), while relatively little is known about intensive margin effects, such as absenteeism. Understanding changes in workplace absence is particularly important given its harmful impacts on productivity (Grinza & Rycx, 2020; Herrmann & Rockoff, 2012) and staffing continuity (Hoey et al., 2023). The CDC estimated that productivity losses linked to all workplace absenteeism cost firms \$225.8 billion in 2015 (CDC Foundation, 2015). Scheduling challenges and higher workloads for covering staff contribute to decreased morale, increased turnover, and potential safety risks (Swart,

¹For a comprehensive review of health outcomes associated with marijuana use, see Anderson and Rees (2023).

2010). These staffing disruptions also have downstream costs across sectors: teacher absences reduce student performance (Herrmann & Rockoff, 2012), and primary care physician absences reduce pediatric patients’ receipt of preventive care (Agha et al., 2025). In broader labor contexts, any legalization-induced increase in absenteeism could likewise erode workforce productivity, increase reliance on temporary staffing, and impair organizational efficiency.

Existing research on the effects of cannabis use on workplace absenteeism is predominantly correlational in nature and remains inconclusive and inconsistent. Studies have reported lower absence (Ullman, 2017), higher absence (Austin et al., 2020; Déguilhem et al., 2022; Morgan et al., 2022; Yang et al., 2024), and no significant effects (Abouk et al., 2021; Lund et al., 2019; van Ours & Williams, 2015). Most of this prior work has relied on correlational designs, or on identification strategies that may limit causal interpretation (Austin et al., 2020; Moughan & Katz, 2024). While Abouk et al. (2021) provide a causal estimate in an auxiliary analysis, their absence measure is limited to absences due to “own illness,” which only accounts for about 22% of total absences in the 2002-2023 CPS ASEC data. In addition, their analysis is restricted to older working adults aged 40–62. It is also necessary to differentiate impacts of medical marijuana legalization, such as those found by Ullman (2017), from those of RMLs. RMLs lead to increased marijuana use among younger, healthier populations than MMLs, and the physical and mental health impacts for these populations differ in response (Borbely et al., 2022; Hollingsworth et al., 2022).

This paper advances the workplace absence literature by estimating the causal effect of recreational marijuana laws on overall absenteeism. Specifically, I estimate the effect of the onset of legal recreational sales using a generalized difference-in-differences event study framework, exploiting the staggered opening of recreational dispensaries across U.S. states over time. All analyses use modern difference-in-differences estimators designed to account for dynamic and heterogeneous treatment effects under staggered policy implementation (Callaway & Sant’Anna, 2021; Sun & Abraham, 2021). The analysis uses state-level panel data on marijuana use from the 2002-2019 National Survey on Drug Use and Health (NSDUH), individual-level repeated cross-sectional data on workplace absence from the 2002-2023 Current Population Survey (CPS), and individual-level repeated cross-sectional data on work-related health outcomes from the 2002-2023 Behavioral Risk Factor Surveillance Survey (BRFSS).

The main contribution of this paper is to provide new causal evidence that retail implementation of recreational marijuana legalization raises short-term workplace absence. The estimates show that the opening of recreational dispensaries leads to a 0.45 percentage point increase in the probability of prior-week absence, a 14.5% increase relative to the control group mean of 3.1%. Several mechanisms could contribute to this increase, including motivational deficits and other cognitive effects of marijuana use. To explore a potential health-related channel, I examine self-reported mental health outcomes. These complementary results show that RMLs lead to increases in self-reported mental health problems that disrupt usual activity, suggesting that deteriorations in mental health following legalization may partially explain the rise in absenteeism.

A key feature of the study design is to distinguish statutory legalization from the onset of legal retail sales. Because retail markets often open well after law passage (Anderson & Rees, 2023; Dave et al., 2025), indicators keyed to statutory dates can attenuate effects if behavior responds primarily to commercialization and access after dispensaries open (Anderson & Rees, 2014). Consistent with this interpretation, my results show that the absenteeism effect appears only when treatment is timed to legal sales onset, while estimates timed to law passage are small and statistically indistinguishable from zero (see Figure 3).

Beyond these main results, the analysis contributes to the existing literature studying marijuana legalization and work absence in three ways. First, many of the aforementioned observational studies are from medical literature and do not use difference-in-differences or other quasi-experimental designs (Austin et al., 2020; Moughan & Katz, 2024). Several do not engage any causal framework (Morgan et al., 2022; Yang et al., 2024). In this paper I use modern difference-in-differences techniques for all analyses, assessing pre-trends and dynamic effects within a robust causal framework. Second, recent studies indicate that traditional TWFE models may be subject to bias when applied to settings with staggered policy adoption (Goodman-Bacon, 2021; Roth et al., 2023). While prior studies in the economics literature rely on traditional TWFE models (Chakraborty et al., 2021; Ullman, 2017) this study applies more recent methodological advancements that address these potential sources of bias. Third, by leveraging the 2002-2023 CPS and BRFSS datasets, the analysis incorporates several more years of post-treatment data than most prior work. Between January 2021 and March 2023, nine additional states opened recreational marijuana dispensaries. Incorporating these additional treated states and years of data into the analysis may facilitate a more comprehensive understanding of RML policy effects.

This paper also contributes to a broader literature on the public health and labor market consequences of marijuana legalization. Existing research examines outcomes salient to policymakers, including traffic fatalities, workplace injuries, crime, and mental health problems (Anderson & Rees, 2023). The findings in this study further inform the public health debate by linking retail market access to workplace absenteeism and to deteriorations in self-reported mental health. Additionally, these findings contribute to the small but growing body of literature on labor market outcomes of marijuana legalization. Prior studies mostly focus on extensive margin outcomes such as employment (Anderson & Rees, 2023; Chakraborty et al., 2021; Jiang & Miller, 2022; Pan, 2025). Only a handful of papers have looked at the intensive margin, including recent work by Dave et al. (2025). This paper adds to that literature by examining a different measure of the intensive margin - workplace absence - and finds significant effects following legalization.

The structure of this paper is as follows: Section 2 provides background on the relationship between marijuana laws and workplace absence, as well as a review of the relevant literature. Section 3 outlines the data, specifying the measures used from publicly available datasets. Section 4 delineates the statistical methods employed in this study. The main findings are detailed in Section 5, with heterogeneity analysis and robustness checks. Section 6 presents supplementary analyses

investigating potential mechanisms behind the observed effects. Section 7 concludes with a discussion of the implications of my findings in context.

2 Background and Related Literature

Several studies have explored the relationship between marijuana policy and workplace absenteeism, particularly in the context of medical marijuana laws or self-reported marijuana use. However, relatively little research has examined this question using recreational marijuana law variation or modern causal inference methods. This section summarizes the existing literature on marijuana and work outcomes, with particular attention to studies of absenteeism. Several plausible mechanisms that may connect cannabis use to short-term work absence are also outlined.

2.1 Potential Mechanisms Linking Marijuana Use to Work Absence

Marijuana use may affect work-related outcomes through the drug’s impacts on mental health and brain functioning. Clinical research shows that marijuana has negative effects on cognition, particularly during acute intoxication (Caulkins et al., 2015; Kroon et al., 2021; Memedovich et al., 2018; Testai et al., 2022). Recent reviews of psychiatric, medical, and neuroscience literature have also found sufficient evidence to establish chronic cognitive effects in heavy users, including impairments in global cognition, dopamine regulation, learning, memory, and attentional control (Hoch et al., 2025; Kroon et al., 2021; Memedovich et al., 2018). Such cognitive effects, whether acute or chronic, may make work more challenging and taking time off more agreeable.

Broad bodies of evidence from economic and medical literature have connected marijuana use with poor mental health (Memedovich et al., 2018; Petrilli et al., 2022; Rup et al., 2021; van Ours & Williams, 2012). Although many consumers perceive self-medication with cannabis to help manage their symptoms (Rup et al., 2021), reviews of current research continue to raise concerns that marijuana may ultimately exacerbate mental health problems, particularly symptoms of psychosis, anxiety, mania, and depression (Hines et al., 2020; Kroon et al., 2021; Memedovich et al., 2018; National Academies of Sciences, Engineering, and Medicine et al., 2017; Petrilli et al., 2022). With regular use, risk is heightened for developing cannabis use disorder, which is further associated with increased risk of mental health problems and neurological harms (Hoch et al., 2025; Petrilli et al., 2022). Heavy use is linked to reduced motivation and changes in reward processing, which may translate into reduced desire to work or engage socially (Hoch et al., 2025; Lawn et al., 2016; Skumlien et al., 2023; Volkow et al., 2016). Alongside this, declines in mental health are significantly associated with increased work absenteeism, and the association appears larger than that for physical health problems (Bryan et al., 2021).

Emerging patterns in the economics literature indicate that mental health consequences affect consumers differently based on age group and other factors (Borbely et al., 2022; van Ours & Williams, 2012). Accordingly, behavioral effects of marijuana legalization on workplace absenteeism

may also differ between groups. In older adults who use medical marijuana, there is evidence of improved psychological health related to cannabis use. In a recent study using generalized DiD methodology, Drake et al. (2025) found that medical marijuana legalization led to a decrease of 3.77 percentage points in poor mental health days over the past month as self-reported by older adults aged 65 and above in New York. Younger recreational users do not appear to gain the same benefits. Borbely et al. (2022) showed heterogeneous effects in a DiD analysis with BRFSS data, finding that MMLs contribute to improved mental health for adults aged 60 and above, particularly those with chronic health conditions; whereas RMLs contribute to negative mental health among people under age 35.

2.2 Self-Reported Marijuana Use and Work Absence

Austin et al. (2020) examine the association between marijuana use and workplace absenteeism using data from the 2006–2007 NSDUH. The authors find that self-reported use of both marijuana and nonmedical use of prescription drugs are positively associated with self-reported absenteeism. While the authors also report IV specifications, the exclusion restriction cannot be tested directly and is therefore difficult to assess for their proposed instruments. Moreover, because the analysis predates recent legalization and dispensary expansion, it cannot leverage policy-driven changes in marijuana access for identification.

There are additional correlational studies, including Moughan and Katz (2024), who investigate the relationship between frequency of cannabis use, legalization, and employee sickness absence using data from the 2019-2020 wave of the Panel Study of Income Dynamics. They regress self-reported workplace absence on a three-way interaction between the legal status of marijuana, a respondent’s frequency of marijuana use, and a respondent’s depression status. They find that higher frequency use among more depressed workers is associated with fewer absences. However, because marijuana use is unlikely to be exogenous in their framework, causal interpretation is limited. In addition, legalization is captured by a single indicator that groups together legalization and decriminalization, which may mask meaningful differences in policy intensity and access (Anderson & Rees, 2014).

Morgan et al. (2022) use the 2018 NSDUH to study the relationship between substance use and workplace absenteeism. Using a negative binomial regression model, they find that self-reported marijuana use is significantly associated with increased workplace absenteeism. However, the authors do not attempt to identify causal effects, as both the exposure and outcome measures are drawn from a single cross-sectional survey. Consequently, the estimates are subject to standard endogeneity concerns, such as reverse causality and omitted variable bias. This limits causal interpretation of the estimated associations.

2.3 Medical Marijuana Law and Work Absence

Turning specifically to medical marijuana laws, Ullman (2017) analyzes CPS ASEC data from 1992 to 2012 to show that state-level workplace sickness absences decrease by 8.3% relative to

the pre-treatment mean after medical marijuana law (MML) adoption. However, this result may not generalize to recreational marijuana policies, because the effects of medical marijuana laws have been shown to differ substantially from recreational laws. For example, Hollingsworth et al. (2022) show that medical marijuana laws increase prior-month adult marijuana use by 5% and have negligible effects on adolescent use, whereas recreational laws generate a 25% increase for adults and a 10% increase in adolescent use. MMLs require consumers to obtain medical authorization based on a qualifying health condition, and this population is more often using marijuana to manage problematic symptoms compared to recreational users, who are generally healthier (Borbely et al., 2022). Correspondingly, medical marijuana use may enable those with severe pain to go to work when they otherwise could not, while recreational consumption may have negative effects on the consistency of working behavior in physically healthy individuals, or in those with risk factors such as predisposition to mental health problems.

2.4 Recreational Legalization and Work Absence Among Older Workers

Abouk et al. (2021) is among the few causal studies to directly examine the relationship between recreational marijuana laws and workplace absenteeism. My analysis differs from theirs in two key respects. First, they study the effect of law passage on older workers aged 40–62, whereas this paper studies the effect of retail market access on a broader working-age population (18–64). This includes younger workers, for whom evidence suggests differential effects to recreational legalization, including adverse mental health responses (Borbely et al., 2022). The older working population studied by Abouk et al. (2021) shows reductions in work-limiting disability and improvements in physical work capacity, suggesting that their marijuana use targets medical benefits even after RMLs.

Second, although the primary focus for Abouk et al. (2021) is the effect of legalization on workers’ compensation claims and physical work capacity, they also conduct an auxiliary analysis of a one-year lagged RML indicator on work absences due to “own health” as reported in the CPS. They find that RMLs increase “own-illness” absences by about 0.06 percentage points, but the estimate is small and not statistically significant. This is an important early contribution, but the auxiliary absence analysis is necessarily narrow in scope. In the CPS, there are ten categories of workplace absence, with “own health” comprising only about 22% of non-layoff absences from 2002–2023. Several of the other categories (such as “vacation/personal days” and “other”) are plausibly more relevant margins if the primary research question is the effect of recreational marijuana policy on work absenteeism broadly. Appendix Table A2 lists all absence categories used in this study.

2.5 Recreational Marijuana Law and Other Work Outcomes

Prior work has examined the labor market consequences of recreational marijuana legalization. Dave et al. (2022) find little to no evidence that recreational marijuana laws negatively affect labor market outcomes for working-age individuals, and instead document modest gains in agricultural employment following legalization. Their study does not examine workplace absenteeism; however,

they do analyze labor supply on the intensive margin. The authors estimate the effect of RMLs on usual weekly hours worked using CPS data and find no evidence of significant changes among the overall workforce following policy adoption (Dave et al., 2022).

Two related studies have explored the relationship between recreational marijuana law and workplace outcomes; both are case studies of particular states and neither examines workplace absenteeism. Jiang and Miller (2022) use Census data from 2000 to 2019 to investigate the effects of recreational marijuana law passage in Washington and Colorado. Using a synthetic control approach, they show that RML passage leads to a decrease in weekly wages in the retail and agricultural sectors. Chakraborty et al. (2021) study the effects of recreational marijuana law passage on labor market outcomes in Colorado. Using county-level data from 2011-2018 in a difference-in-differences framework, they find that recreational dispensary openings increased local employment.

3 Data

3.1 National Survey of Drug Use and Health (NSDUH)

I begin by examining changes in marijuana use among adults ages 18+ using data from the 2002–2019 National Survey on Drug Use and Health (NSDUH). I follow the literature for variable definitions with this data (Dave et al., 2025; Sabia et al., 2024). While individual-level geographic identifiers are restricted, state-by-year averages are publicly available for this period. The survey collects data on marijuana use in each wave, with results reported over rolling two-year intervals. Consistent with SAMHSA’s guidance (Substance Abuse and Mental Health Services Administration, 2025), I restrict attention to the harmonized 2002–2019 NSDUH series.²

The use of overlapping two-year survey periods complicates the construction of state-level marijuana policy indicators that align precisely with actual policy timing. To address this, I follow the approach in Choi et al. (2019, p. 312), defining policy exposure as the proportion of each two-year period during which the policy was in effect. These indicators range from 0 (no exposure) to 1 (full exposure). For instance, if a state enacted the policy in month 16 of the 24-month period, its value would be coded as 0.33.

Marijuana prevalence rates are based on self-reported data of how many days in the last month each respondent used marijuana or hashish. Marijuana users are identified as the respondents reporting one day of marijuana use in the past month or more. Approximately 7.5 percent of all adult respondents met this criteria for marijuana use between 2002 and 2019. Some age groups have higher prevalence rates, with the highest rate of marijuana use at approximately 19 percent among those aged 18 to 25, contrasting the rate of 5.5 percent among respondents aged 26 and

²Beginning in late 2020, NSDUH introduced web-based interviewing and a revised variable crosswalk, creating a break in comparability with earlier years. SAMHSA advises that data from 2020 onward cannot be pooled with 2002–2019, and no state-level estimates were released for 2019–2020 due to pandemic disruptions. The analysis therefore uses the last continuous set of state-consistent data available, 2002–2019.

older. Prevalence rates overall steadily increased across the sample period, from 6 percent of all adult respondents in 2002 up to 11 percent in 2019.

3.2 Current Population Survey (CPS ASEC)

The primary analysis uses data from the 2002-2023 Current Population Survey Annual Social and Economic Supplement (ASEC). All of the data in this sample was observed between February and April of each year, permitting temporally precise policy variables. The CPS instructions note that any estimation performed on this sample must use their supplied inverse probability weight to generate population-representative results.³ Accordingly, all estimation on this dataset is weighted by ASEC sampling weights (Abouk et al., 2021; Ullman, 2017).

The analysis is restricted to employed observations, which is the effective universe of the absent question. The outcome measure, $Absent_{ist}$, is generated from a survey question which asks whether or not the respondent was absent from work in the past week. Respondents are also asked why they were absent from work in a follow-up question, with ten possible responses. The categories and corresponding absence counts are summarized in Appendix Table A2. I exclude respondents who report “maternity or paternity leave,” which comprise 6.23% of total absences, because this typically requires advanced notice and a formal process. I also exclude respondents who report “labor disputes” and “civic or military duty”, equal to 0.32% of total absences. The remaining categories are all included in $Absent_{ist}$.

Furthermore, to attenuate potential confounding variation related to childbirth and childcare availability, which may vary between treated and control states, I restrict the sample to exclude observations with newborn infants less than one year old. This group comprises 3.2% of the employed sample. After all stratification described in this section, 1,789,100 out of 2,011,953 employed observations aged 18-64 remain in the sample (88.9%), with 52,465 reported workplace absences. Table 1 provides descriptive statistics for the sample.

3.3 Behavioral Risk Factor Surveillance Survey (BRFSS)

To explore potential mechanisms, I use all available data from the Behavioral Risk Factor Surveillance System (BRFSS) for the years 2002–2023. While mechanisms operating through motivation or cognition are difficult to assess directly, these data allow me to examine health-related channels. The BRFSS provides a cross-sectional dataset on the health behaviors of a large sample of nationally representative individuals. As in prior analyses, the sample is restricted to employed individuals aged 18–64.

³Weighting guidance and documentation is available from IPUMS (2025).

Table 1: Descriptive Statistics for Outcome Variables

	All States		Treated States		Control States	
	Mean	SD	Mean	SD	Mean	SD
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: NSDUH (2002-2019)						
<i>Used marijuana in the last 30 days</i>						
Rate, Ages 18+	7.5%	(2.6%)	9.8%	(3.1%)	6.8%	(1.9%)
Ages 18-25	18.7%	(4.2%)	22.2%	(4.6%)	17.9%	(3.6%)
Ages 25+	5.6%	(2.4%)	7.6%	(3.1%)	5.0%	(1.8%)
State-year Observations	867		136		731	
Number of States	51		8		43	
Panel B: CPS ASEC (2002-2023)						
<i>Absent from work last week</i>						
Rate, Ages 18-64	3.00%	(0.10%)	3.23%	(1.17%)	2.85%	(0.85%)
Individual Observations	1,787,708		775,831		1,011,877	
Number of States	51		20		31	
Panel C: BRFSS (2002-2023)						
<i>Physical and/or mental health problems disrupted usual activity</i>						
Days, Ages 18-64	1.26	(0.34)	1.31	(0.35)	1.21	(0.35)
<i>Days disrupted exclusively by mental health problems</i>						
Days, Ages 18-64	0.28	(0.16)	0.31	(0.17)	0.27	(0.15)
<i>Days disrupted exclusively by physical health problems</i>						
Days, Ages 18-64	0.40	(0.09)	0.41	(0.08)	0.39	(0.09)
State-year Observations	1,122		434		688	
Number of States	51		20		31	

Notes: This table presents the average characteristics across all states (columns 1–2), treated states (columns 3–4), and control states (columns 5–6) utilized in the baseline analysis. It reports both mean values and standard deviations, with observations weighted according to relevant state population using SEER data. The initial two columns encompass all states, while columns 3–4 focus on the treated states that enacted a Recreational Marijuana Law (RML) and opened recreational dispensaries. The final two columns pertain to control states that did not open recreational dispensaries prior to the conclusion of the sample period.

Every year, survey respondents are asked several questions about their recent mental and physical health. I construct three outcomes from these questions: the number of days in the past month that poor health prevented usual activities (such as work); the number of such days attributable to mental health problems only; and the number attributable to physical health problems only. To focus on population-level impacts and reduce measurement error in self-reported health outcomes, I aggregate the data to the state-year level prior to analysis (Borbely et al., 2022).

3.4 Recreational Marijuana Law Dates

I rely on the effective dates compiled by Anderson and Rees (2023), supplemented with independent research into further legislative changes. The complete set of implementation dates is shown in Appendix Table A1, including the dates when legal recreational markets opened.

4 Methods

I leverage variation in the timing of recreational marijuana law implementation across states to estimate their impact on workplace absence rates. Drawing from existing literature on recreational marijuana laws (Dave et al., 2025; Hollingsworth et al., 2022; Sabia et al., 2024), the empirical strategy comprises a generalized difference-in-differences framework suited to staggered policy implementation. Recent contributions to the econometrics literature highlight that TWFE estimates may be biased in settings with staggered treatment timing (Goodman-Bacon, 2021), which can also distort standard event study results used to assess pre-treatment trends (Callaway & Sant’Anna, 2021; Sun & Abraham, 2021). Accordingly, my primary static and dynamic estimates are obtained using the interaction weighted estimator proposed by Sun and Abraham (2021), following recent applied work in staggered adoption settings (Chlond & Germeshausen, 2025; Encinosa & Dor, 2025; Kose et al., 2024).

4.1 Empirical model

I implement the interaction weighted event study estimator using the following specification:

$$Y_{st} = \beta_0 + \sum_{e \notin C} \sum_{\substack{l=-5 \\ l \neq -1}}^5 \beta_{e,l} (\mathbb{1}\{E_s = e\} \times RMD_{st}^l) + \beta_2 MML_{st} + \beta_3 MJDecrim_{st} + \dots \\ \dots + X_{st}'\beta_4 + \alpha_s + \gamma_t + \epsilon_{st} \quad (1)$$

where Y_{st} denotes a given outcome observed in state s during year t . These outcomes include the prevalence of prior-month marijuana use, prior-week workplace absence, and the number of prior-month days suffering physical or mental health problems that disrupt usual activity. When

using the CPS ASEC, I estimate a micro-level analogue of Equation (1), adapted for individual-level data.⁴

The treatment indicator of interest, $RMD_{s,t}^l$, is an event-time dummy equal to 1 if $\{t - E_s = l\}$, where E_s is the first year in which state s began legal recreational marijuana sales. l indexes relative years to treatment, representing the duration in years between t and E_s . C denotes the control group, the set of all never-treated observations, thereby avoiding bias from comparing early adopters to later adopters. $\mathbf{1}\{E_s = e\}$ is an indicator function where e indexes each treated cohort by year of treatment. The final interaction weighted estimates for each relative time period are constructed by weighting the recovered $\hat{\beta}_{e,l}$ coefficients by cohort sample shares (Cullen & Pakzad-Hurson, 2023; Sun & Abraham, 2021). All estimation is performed using the Stata package provided by Sun & Abraham.

To distinguish the estimated effects from earlier cannabis policy changes, I also control for medical marijuana laws and marijuana decriminalization laws with $MML_{s,t}$ and $MJDecrim_{s,t}$ (Dave et al., 2025; Mathur & Ruhm, 2023; Sabia et al., 2024). The recovered coefficients, the final set of estimates $\hat{\beta}_t^{IW}$, therefore capture the marginal conditional average treatment effects of recreational marijuana dispensaries opening over and above the effects induced by medical law passage and marijuana decriminalization laws.

The vector X_{st} includes a set of time-varying, state-level covariates drawn from data sources including the ACS, FRED, and CDC. These variables capture shifts in state population characteristics over time, and include demographics controls (age, gender, race, ethnicity, education level); macroeconomic controls (the real state GDP, the real median household income); social welfare policies (whether or not the state adopted mandatory paid family leave laws, whether or not the state expanded Medicaid); and substance use controls (whether the state has a mandatory prescription monitoring drug program, the real state beer tax, the real state cigarette tax). α_s is a vector of time-invariant state effects and γ_t is a vector of state-invariant time effects. I report estimates under increasingly rich specifications, starting from a baseline with state and year fixed effects only, then incrementally adding the time-varying state-level covariates in X_{st} .

I estimate event-study specifications for all outcomes using Equation (1). To summarize the post-treatment effects with a single scalar, I also report an average post-treatment effect constructed from the interaction-weighted event-time coefficients, as in Cullen and Pakzad-Hurson (2023) and Chlond and Germeshausen (2025). The summary measure averages the post-treatment event-time coefficients using weights proportional to the number of treated observations available at each event time. This mirrors the weighting logic emphasized by Deb et al. (2025), preventing later lags with limited support from receiving the same weight as earlier post-treatment periods with substantially greater support.

Treatment effects are identified from cross-state variation in the timing of recreational marijuana

⁴This specification adds Z_{it} , a vector of individual demographic controls for gender, race, age, and educational attainment; these replace the analogous state-level demographic controls in X_{st} . The model is otherwise identical.

legalization and the subsequent onset of legal retail sales. A causal interpretation of the difference-in-differences estimates requires that, absent treatment, outcomes in treated and comparison states would have evolved in parallel. It also requires no anticipatory behavior: workplace absence should not respond prior to the onset of legal sales. These assumptions may be violated if treated states were already on different pre-policy trajectories, or if RMLs were enacted in response to unobserved time-varying factors correlated with workplace absence.

To assess the plausibility of these assumptions, I examine the Sun and Abraham (2021) event study lead coefficients from Equation (1) as a diagnostic for differential pre-treatment trends and anticipatory responses. A further identifying assumption is no interference across states. I probe potential geographic spillovers in Figure B2 by excluding bordering states from the comparison group, and the resulting dynamic estimates are qualitatively similar to the baseline specification.

4.2 Alternative Estimation

As a robustness check, I re-estimate Equation (1) for all outcomes using the group-time average treatment effect framework of Callaway and Sant’Anna (2021). I use their doubly robust inverse probability weighted procedure, which identifies group-time effects $ATT(g, t)$ by combining an outcome model with propensity score reweighting and then aggregates these effects to obtain an overall ATT . This serves as a useful contrast to the Sun and Abraham (2021) approach, because they differ both in their estimation procedure and in how dynamic effects are aggregated into a scalar summary (Baker et al., 2022). I report the resulting ATT estimates in the main tables and the event study results in Appendix Figure B1. All results are qualitatively similar between methods.

5 Results

This section proceeds in four parts. First, I present evidence that adult marijuana use rises following state adoption of RMLs with operational recreational dispensaries. Second, I show that these RMLs lead to increases in workplace absence. Third, I subject these findings to rigorous robustness checks to confirm that they are not driven by sample composition, alternative model specifications, or spurious policy timing. Fourth, I examine heterogeneity in treatment effects across absence types and by respondent income. All standard errors are corrected for clustering at the state level (Bertrand et al., 2004).

5.1 Marijuana use

I begin by estimating the impact of RMLs with operational recreational dispensaries on adult marijuana consumption. As shown in Table 2, ATT estimates derived from Equation (1) indicate a statistically significant increase in usage. Estimates exhibit minimal sensitivity to the inclusion of additional covariates, with similar results observed across all three specifications. Panel A displays estimates for any marijuana use in the past 30 days among adults aged 18 and older.

Table 2: ATT estimation for effect of RML with recreational dispensaries on prevalence of prior-month marijuana use (NSDUH, 2002-2019)

	(1)	(2)	(3)	(4)
	No Covariates	Add Demographics Controls ¹	Add Macro & Social Policy Controls ²	Callaway Sant’Anna
Panel A: All 18+				
<i>RML with dispensaries</i>	0.0401*** (0.0037)	0.0375*** (0.0046)	0.0373*** (0.0046)	0.0340*** (0.0133)
Control Mean	0.0713	0.0713	0.0713	0.0713
State-year Observations	867	867	867	867
Panel B: Age Heterogeneity				
Ages 18-25				
<i>RML with dispensaries</i>	0.0447*** (0.0102)	0.0413*** (0.0119)	0.0432*** (0.0107)	0.0425** (0.0182)
Control Mean	0.1874	0.1874	0.1874	0.1874
State-year Observations	867	867	867	867
Ages 26+				
<i>RML with dispensaries</i>	0.0404*** (0.0083)	0.0371*** (0.0086)	0.0377*** (0.0083)	0.0329*** (0.0126)
Control Mean	0.0543	0.0543	0.0543	0.0543
State-year Observations	867	867	867	867
Demographic Controls	No	Yes	Yes	No
Social Welfare Policies	No	No	Yes	No
Substance Use Controls	No	No	Yes	No
Model	S&A	S&A	S&A	CS

* $p < 0.10$, ** $p < .05$, *** $p < .01$.

Notes: Each column represents a separate regression. Results are derived using Sun and Abraham’s (2021) dynamic treatment effect estimator to account for timing-induced bias. Data are from the 2002-2019 NSDUH. All models include state and year fixed effects. Standard errors, clustered at the state level, are in parentheses.

¹ Demographic controls encompass age, gender, race, ethnicity, and education level.

² Macroeconomic and social policy controls include the real median household income, the real state GDP, a state-level indicator for Medicaid expansion under the Affordable Care Act, mandatory prescription-monitoring drug laws, the real beer tax per gallon, and the real cigarette tax per pack.

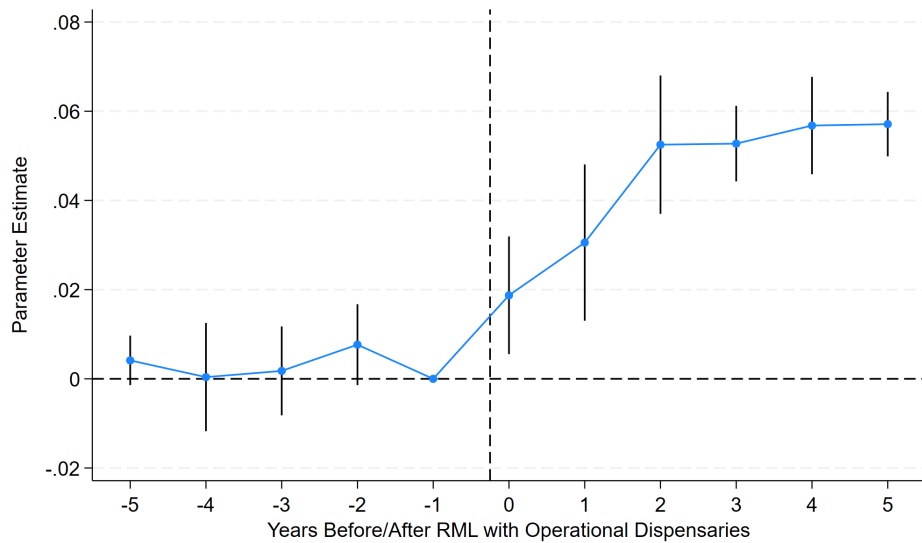
In the preferred model (column 3), which includes all covariates, the onset of legal retail sales is associated with a 3.73 percentage point rise in the probability of recent marijuana use. This roughly corresponds to a 50% increase in the likelihood of adult marijuana use compared to the baseline mean in states that adopted RMLs.

The magnitude of this effect aligns closely with many estimates from recent literature, which mostly use policy instruments that do not distinguish between RML passage and retail dispensary operations. For example, relying on law enactment as the policy instrument, Dave et al. (2023), Hollingsworth et al. (2022), and Sabia et al. (2024) each find that adult past-month marijuana use increased by approximately 50% relative to baseline levels in states adopting legalization, based on 2002–2019 NSDUH data. This is roughly equal to 4 percentage points, comparable to the effect identified in Panel A of Table 2.

Panel B explores age-based heterogeneity, separating the analysis into adults aged 18–25 and those aged 26 and above. I find significant increases in marijuana use for both age groups after recreational dispensaries open. Among 18-25 year olds, I find that RMLs with active dispensaries lead to a 4.32-percentage point increase in the prevalence of prior-month marijuana use. I find a similar increase of 3.77-percentage points for adults aged 26 and over. These effects constitute a 23 and 69 percent increase over the control means, respectively.

Figure 1 presents the event study estimates for “any marijuana use” in the prior month among adults, derived using the interaction weighted estimator proposed by Sun and Abraham (2021).

Figure 1: Sun & Abraham Event Study, Effect of RML with recreational dispensaries on prevalence of prior-month marijuana use (NSDUH, 2002-2019)



Notes: The graph contains coefficient estimates with 95% confidence intervals for years before and after the change in RML, derived using Sun and Abraham’s (2021) dynamic treatment effect estimator to account for timing-induced bias. The regression controls for all covariates in X_{st} , state and time fixed effects. Standard errors are clustered at the state level. Data are from the 2002-2019 NSDUH and include all observations aged 18+.

There is no significant evidence of differential trends in adult marijuana use between treated and control states before recreational dispensaries opened. All pre-period estimates are statistically indistinguishable from zero ($p > 0.05$). This pattern supports the identifying assumption of parallel pre-treatment trends in the baseline difference-in-differences framework.

After recreational dispensaries open, the prevalence of adult marijuana use increases significantly in treated states during the first year, then increases again over the next two years to approximately 5.5 percentage points beyond the pre-treatment mean. The effect plateaus there, but remains elevated for at least three more years. These findings are consistent with a gradual expansion in marijuana availability following the implementation of recreational dispensaries.

5.2 Rates of work absence

Next, I turn to repeated cross-sectional data from the 2002-2023 CPS ASEC. I find causal evidence that recreational marijuana laws with operational dispensaries lead to an increase in workplace absenteeism. Table 3 below contains the aggregated results from Equation (1), estimates of the change in the probability of an employed observation reporting prior-week workplace absenteeism in states with operational recreational marijuana dispensaries versus those without.

Table 3: ATT Estimates for effects of RML with recreational dispensaries on workplace absenteeism (CPS ASEC, 2002-2023)

	(1)	(2)	(3)	(4)
	No Covariates	Add Demographics Controls	Add Macro & Social Policy Controls	Callaway Sant’Anna
Absent from Work				
<i>RML with dispensaries</i>	0.0044*** (0.0012)	0.0047*** (0.0012)	0.0045*** (0.0012)	0.0047*** (0.0014)
Control Mean	0.0310	0.0310	0.0310	0.0310
Individual Observations	1,774,284	1,774,284	1,774,284	1,774,284
Demographic Controls	No	Yes	Yes	No
Social Welfare Policies	No	No	Yes	No
Macroeconomic Controls	No	No	Yes	No
Model	S&A	S&A	S&A	CS

* $p < 0.10$, ** $p < .05$, *** $p < .01$.

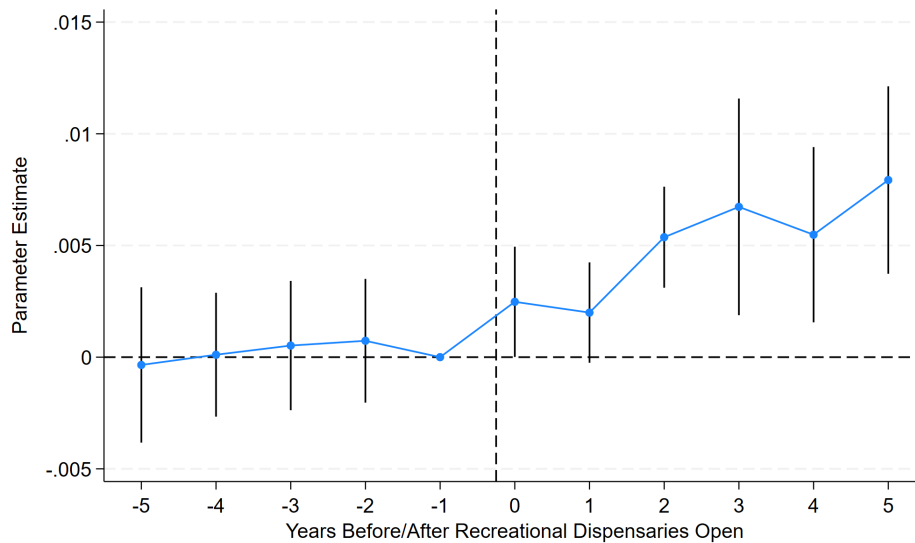
Notes: Each column represents a separate regression. Results are derived using Sun and Abraham’s (2021) dynamic treatment effect estimator to account for timing-induced bias. Data are from the 2002-2023 CPS ASEC. All models include state and year fixed effects. Standard errors, clustered at the state level, are in parentheses.

Column (3) reports the main specification. Generally, among adults aged 18-64, I find that state adoption of RMLs with active retail dispensaries leads to a 0.45 percentage point increase in the probability that an employed person will be absent from work in the prior week. This effect represents an approximate 14.5% increase over the control group mean absence rate. In columns (1) and (2), I assess the integrity of the main specification by running the model without the full set of controls. The results are robust to this exclusion, with minimal changes in coefficient size or standard error. This suggests that the main findings are not sensitive to covariate specification.

I next use Equation (1) to evaluate pre-treatment trends and investigate how effects evolve over time, using an event study approach estimated with recent methodology proposed by Sun and Abraham (2021). The results are presented in Figure 2. Consistent with the identifying assumptions, I find no significant evidence of differential trends in workplace absence between treated and control states in the years preceding the onset of legal recreational sales. All pre-period point estimates for differences between treated and control states are centered near zero. This pattern bolsters the credibility of the parallel trends assumption underlying the identification strategy.

In the year after recreational dispensaries open, there is a sharp increase in the workplace absence rate. There is another increase two years later to around 0.5 percentage points, after which the effect remains elevated for at least three additional years. This pattern is consistent with the gradual increase in marijuana use observed following the expansion of legal access through dispensaries.

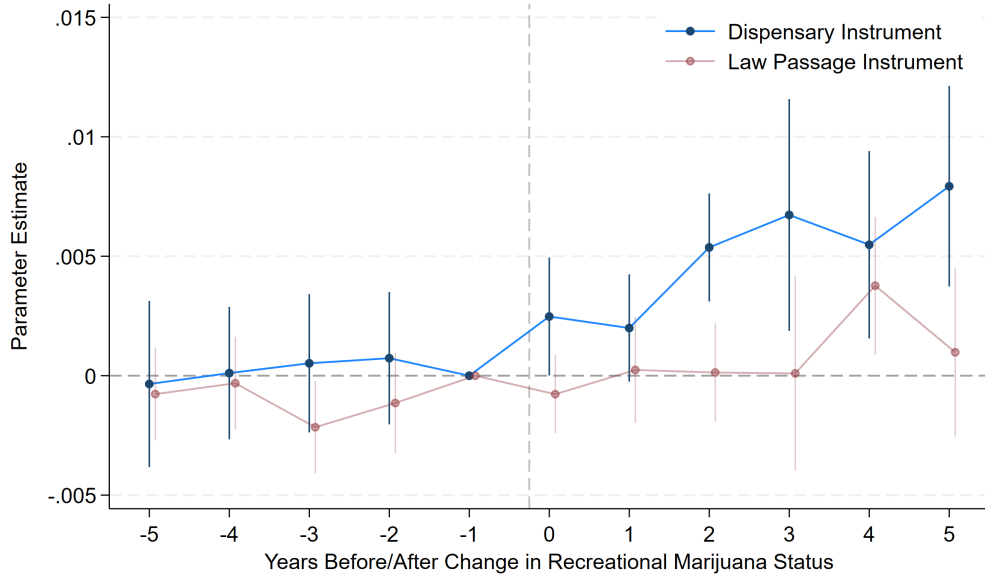
Figure 2: Sun & Abraham Event Study, Effects of RML with recreational dispensaries on workplace absenteeism (CPS ASEC, 2002-2023)



Notes: The graph contains coefficient estimates with 95% confidence intervals for years before and after the change in RML, derived using Sun and Abraham's (2021) dynamic treatment effect estimator to account for timing-induced bias. The regression controls for all covariates, state and time fixed effects, and is weighted by CPS ASEC sampling weights. Standard errors are clustered at the state level. Data are from the 2002-2023 CPS ASEC and include employed observations aged 18-64.

5.3 Policy timing: law passage versus retail sales

Figure 3: Sun & Abraham Event Study, Effects of RML on workplace absenteeism (CPS ASEC, 2002-2023)



Notes: The graph contains coefficient estimates with 95% confidence intervals for years before and after the change in RML, derived using Sun and Abraham’s (2021) dynamic treatment effect estimator to account for timing-induced bias. Each line represents a separate regression. The regressions control for all covariates, state and time fixed effects, and are weighted by CPS ASEC sampling weights. Standard errors are clustered at the state level. Data are from the 2002-2023 CPS ASEC and include all employed observations aged 18-64.

In this section, I assess whether the estimated effects of recreational marijuana policy on workplace absenteeism depend on how the policy’s timing is defined. Figure 3 overlays event-study estimates using RML passage as the treatment indicator (red, transparent line) with this paper’s baseline specification that uses the date legal retail sales begin (blue line). The models are otherwise identical. When treatment is defined by law passage, the post-treatment coefficients are small and statistically indistinguishable from zero. The effect emerges only when defining treatment based on the onset of legal retail sales, suggesting that changes in absenteeism are driven by the availability of commercial retail markets rather than the timing of legislative approval alone.

This pattern is consistent with how recreational markets develop on the ground. In the interim between law passage and the opening of recreational dispensaries, which averages approximately 12.2 months (Dave et al., 2025), legal channels for new consumers remain limited. Home growing projects require significant investments in time, equipment, and expertise and are typically undertaken by a small percentage of consumers who already identify with marijuana use as an important part of their lives (Athey & Newhart, 2025). Before dispensaries open, retail sales are prohibited and there is little evidence of a large-scale expansion in gray-market production or distribution during this interim period in most states (Schlien et al., 2024).

In contrast, the onset of legal retail sales is accompanied by a rapid expansion of legal access through licensed outlets, a broader menu of products and modes of consumption (such as edibles and vape cartridges), greater commercial exposure, and reduced stigma around recreational use (J. Brown et al., 2023; Dave et al., 2025; Dills et al., 2021). These changes plausibly lower search and time costs and make it easier for existing users to increase their frequency of use. If workplace absenteeism responds primarily to these access-driven shifts rather than to the formal legal status of marijuana, then effects may coincide with dispensaries opening rather than law passage.

5.4 Additional Robustness Checks

To further evaluate the reliability of my findings, I perform three supplemental analyses. First, I test the robustness of the main results to the exclusion of parents with preschool-aged children. I estimate Equation (1) for the effect of recreational dispensaries opening on prior-week workplace absence while excluding observations with children under 6 from the sample. I present ATT estimates in Panel A of Table 4 below, and the event study results in Panel A of Figure 4. Estimates remain stable when parents of preschool-aged children are excluded from the sample. The consistency of these results suggests that differences in parental leave or childcare availability are not driving the estimated effects.

Second, I assess the potential influence of the COVID-19 pandemic by re-estimating Equation (1) while excluding data from 2020. The results, presented in Panel B of Table 4 and Figure 4, are highly consistent with the main estimates across both the Sun & Abraham and Callaway Sant’Anna specifications. This suggests that the observed treatment effects are unlikely to be driven by pandemic-related disruptions.

Third, I investigate the possibility that the main results may be driven by spurious correlation by performing a placebo treatment test on Equation (1). I perform 1000 Monte Carlo simulations, generating falsified treatment scenarios by substituting the key policy variable, RMD_{st} , with a randomly generated placebo indicator. In each iteration, treatment status and timing are randomly assigned from uniform distributions, with an expected placebo treatment share of approximately 10%. This matches the observed policy adoption rate in the 2002–2023 CPS sample. I collect the estimated ATT from each simulation, then compile them into the distribution shown in Figure 5.

The placebo test results reinforce the plausibility of the main findings shown in Table 3. The distribution of simulated treatment effects approximates a normal distribution with mean 0, consistent with a null difference-in-differences effect when no actual policy was applied. This finding is consistent with the parallel trends assumption. Next, I overlay the main estimate from Equation (1), capturing the impact of recreational marijuana dispensaries opening on workplace absence rates, onto the placebo distribution. This estimate lies within the upper 1% tail of simulated coefficients, indicating that the observed effects in Table 3 are unlikely to have arisen by chance.

Table 4: ATT Estimates for Sensitivity of Main Analysis (CPS ASEC, 2002-2023)

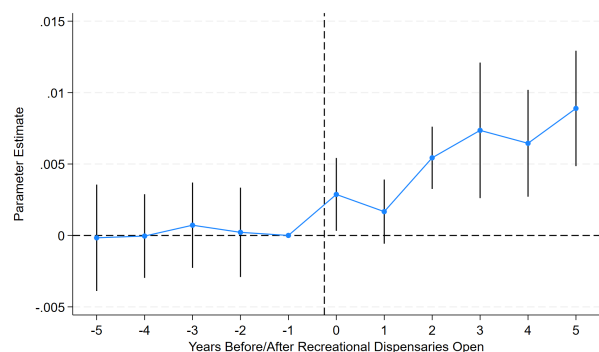
	(1)	(2)	(3)	(4)
	No Covariates	Add Demographics Controls	Add Macro & Social Policy Controls	Callaway Sant'Anna
A: Excluding Parents of Children Under 6				
<i>RML with dispensaries</i>	0.0047*** (0.0012)	0.0047*** (0.0011)	0.0048*** (0.0011)	0.0049*** (0.0015)
Control Mean	0.0307	0.0307	0.0307	0.0307
Individual Observations	1,488,768	1,488,768	1,488,768	1,488,768
Model	S&A	S&A	S&A	CS
B: Excluding 2020				
<i>RML with dispensaries</i>	0.0048*** (0.0012)	0.0048*** (0.0012)	0.0051*** (0.0012)	0.0054*** (0.0014)
Control Mean	0.0292	0.0292	0.0292	0.0292
Individual Observations	1,681,640	1,681,640	1,681,640	1,681,640
Model	S&A	S&A	S&A	CS

* $p < 0.10$, ** $p < .05$, *** $p < .01$.

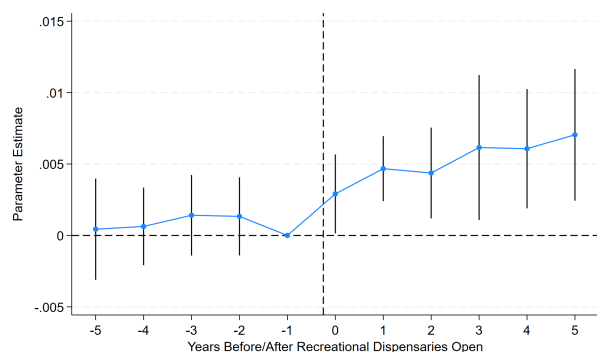
Notes: Each column represents a separate regression. Results are derived from linear probability models estimated through OLS. All regressions control for all covariates, state and time fixed effects, and are weighted by CPS ASEC sampling weights. Standard errors are clustered at the state level. Data are from the 2002-2023 CPS ASEC and include all employed observations aged 18-64.

Figure 4: Sun & Abraham Event Studies, Sensitivity of main outcome (CPS ASEC, 2002-2023)

(a): Excluding Parents with Children Under 6

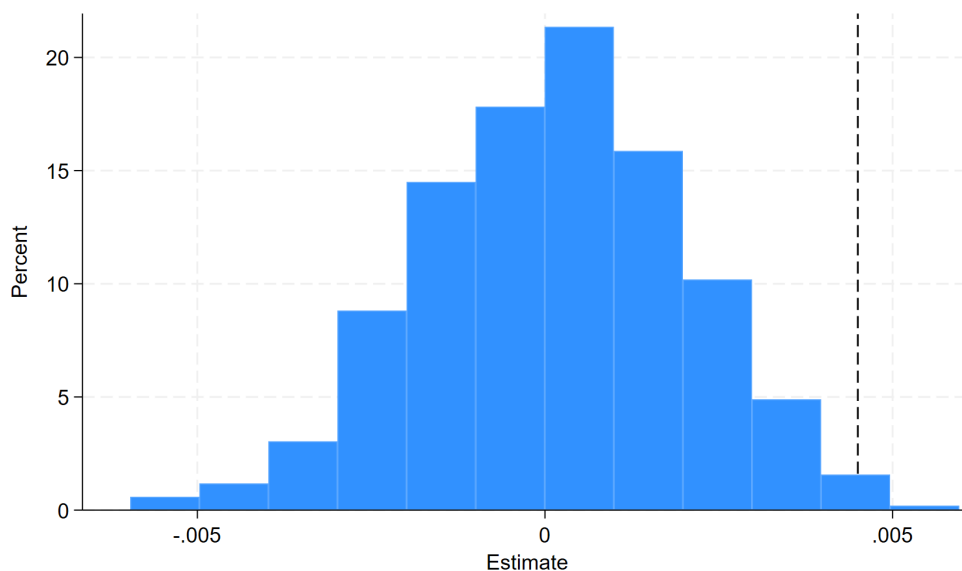


(b): Excluding 2020



Notes: The graphs contain coefficient estimates with 95% confidence intervals for years before and after the change in RML, derived using Sun and Abraham's (2021) dynamic treatment effect estimator to account for timing-induced bias. The regressions control for all covariates, state and time fixed effects, and are weighted by CPS ASEC weights. Standard errors are clustered at the state level. Data are from the 2002-2023 CPS ASEC and includes employed observations aged 18-64.

Figure 5: Placebo Test, Effect of RML with recreational dispensaries on workplace absence (CPS ASEC, 2002-2023)



Notes: The graph plots the density of parameter estimates from 1000 Monte Carlo simulations of equation (1), replacing the main regressor (RMD_{st}) with a randomly generated placebo policy date drawn from a uniform distribution during each iteration. The dashed line represents the main estimate from column (3) of Table 3. Less than 1% of estimates fall to the right. Data are from the 2002-2023 CPS ASEC and includes employed observations aged 18-64.

5.5 Heterogeneity by absence type and respondent income

To shed light on the mechanisms underlying the aggregate increase in work absenteeism, I examine whether the response to recreational dispensaries varies by the reported reason for leave. I re-estimate the main specification separately for each category of absence. Table 5 contains the results, showing significant increases in absences for vacation, illness, and “other,” while weather- and training-related absences are statistically unchanged, consistent with null effects. Increases in vacation/personal and “other” leave account for roughly two-thirds of the overall rise in absences, with sick leave making up the remaining one-third.

While sickness absences may increase due to a health-based effect, the category of absences for vacation or personal time may be interpreted as discretionary. This pattern points toward at least two distinct, but potentially interacting, channels. Increases in sick leave are consistent with a health and work-capacity mechanism, while the rise in vacation or personal leave suggests a preference and motivation channel. After recreational sales begin, frequent marijuana use may alter the relative valuation of work versus leisure, potentially via changes in motivation and reward systems in the brain (Hoch et al., 2025). If the marginal utility of leisure rises, or if work becomes more saliently unpleasant when contrasted with readily available recreational activities, workers may be more willing to take vacation or personal days when opportunities arise (Aguiar et al., 2021).

Table 5: Heterogeneity by Type of Absence (CPS ASEC, 2002-2023)

	Absence Type				
	(1) Vacation or Personal Days	(2) Own Illness	(3) Other	(4) Weather	(5) School or Training
Absent from Work					
<i>RML with dispensaries</i>	0.0018** (0.0008)	0.0014*** (0.0005)	0.0010*** (0.0003)	0.0003 (0.0002)	0.0001 (0.0002)
Control Mean	0.0135	0.0073	0.0067	0.0013	0.0007
Individual Observations	1,746,031	1,735,038	1,734,096	1,724,621	1,723,494
Model	S&A	S&A	S&A	S&A	S&A

* $p < 0.10$, ** $p < .05$, *** $p < .01$.

Notes: Each column represents a separate regression. Results are derived using Sun and Abraham’s (2021) dynamic treatment effect estimator. Each regression controls for all covariates, state and time fixed effects, and is weighted by CPS ASEC sampling weights. Standard errors are clustered at the state level. Data are from the 2002-2023 CPS ASEC and include employed observations aged 18-64.

I next examine whether the impact of RMLs with dispensaries on work absence is concentrated among lower- or higher-earning workers by re-estimating the main specification separately for each quintile of real wage income. Table 6 shows significant increases in absenteeism in the first, second, and fifth income quintiles, while effects in the third and fourth quintiles are small and statistically indistinguishable from zero.

The larger effects observed in the lower income quintiles are particularly consequential given the limited access of low-wage workers to paid leave and other forms of income smoothing. Employer-provided paid sick leave, short-term disability, and other forms of pay protection are substantially less common in low-wage jobs than in higher-wage occupations (Wiatrowski, 2015). As a result, a given increase in work absence is more likely to translate into direct income loss for workers in the bottom quintiles. Moreover, low-wage workplaces often operate with lean staffing and limited slack, so coworkers who cover these absences may do so under higher stress and with greater safety risks (Swart, 2010).

Table 6: Heterogeneity by Income Quintile of Respondent

	Income Quintile				
	(1) First	(2) Second	(3) Third	(4) Fourth	(5) Fifth
Absent from Work					
<i>RML with dispensaries</i>	0.0059*** (0.0019)	0.0068*** (0.0020)	0.0004 (0.0010)	0.0023 (0.0022)	0.0037*** (0.0012)
Control Mean	0.0378	0.0246	0.0237	0.0246	0.0233
Individual Observations	328,437	326,243	328,429	326,101	327,305
Model	S&A	S&A	S&A	S&A	S&A

* $p < 0.10$, ** $p < .05$, *** $p < .01$.

Notes: Each column represents a separate regression. Results are derived using Sun and Abraham's (2021) dynamic treatment effect estimator. Each regression controls for all covariates, state and time fixed effects, and is weighted by CPS ASEC sampling weights. Standard errors are clustered at the state level. Data are from the 2002-2023 CPS ASEC and include employed observations aged 18-64.

6 Changes in mental health as a mechanism

The preceding analysis documents that the onset of legal recreational marijuana sales leads to an elevated incidence of self-reported workplace absence. Multiple mechanisms may contribute to this pattern, including possible effects of amotivational symptoms or cognitive declines subsequent to frequent marijuana use. Although the channels operating through motivation or cognition are difficult to assess directly, the BRFSS provides measures of recent mental and physical health that allow me to examine health-related mechanisms. As with the primary analysis, I focus on individuals aged 18 to 64 who report being employed. I look at the number of days per month, on average, the workforce suffers mental or physical health problems that disrupt normal activity⁵. The use of this measure complements Borbely et al. (2022) by capturing functional limitation, which directly aligns with workplace absence.

Table 7 displays results from that exercise using the same model specifications from the primary analyses, with column (3) containing estimates from the preferred specifications. Panel A reports changes in the number of prior-month days the workforce suffers physical or mental health problems that disrupt regular activity, including working. I find that among adults aged 18-64, opening recreational marijuana dispensaries leads to an additional 0.0714 problem days in the past month, an approximate 6% increase over the control mean of 1.22 days.

Panel B repeats the activity-limitation analysis for a subsample reporting physically unhealthy days but no mentally unhealthy days, isolating a physical-health-only margin. In this group, the estimated effect of dispensaries opening is small and statistically indistinguishable from zero, suggesting that the increase in disrupted-activity days is not driven by physical health shocks. Panel C implements the analogous restriction for respondents reporting mentally unhealthy days but no physically unhealthy days. In contrast, I find a statistically significant increase in disrupted-activity days for this group. Taken together, the results imply that the observed increase in disrupted-activity days following dispensary openings is concentrated in disruptions due to mental health rather than physical health problems.

These findings suggest that recreational marijuana laws with dispensaries may lead to declines in the mental health of the workforce. This could explain a substantial portion of the absence identified in the main analysis. However, as with the primary analysis, these findings are vulnerable to contamination from differential trends in the pre-treatment period. There is also substantial policy timing heterogeneity across the sample period. I address both concerns by estimating Equation (1) on all of these outcomes to conduct event study analyses, estimated using the interaction weighted estimator proposed by Sun and Abraham (2021). This should expunge potential bias arising from problematic comparisons in outcomes between the states which adopted RMLs early in time, versus those that did so much later (Goodman-Bacon, 2021; Sun & Abraham, 2021).

⁵This outcome is generated from a BRFSS survey question asking respondents, “During the past 30 days, for about how many days did poor physical or mental health keep you from doing your usual activities, such as self-care, work, or recreation?”

The results for prior-month days with disruptive health problems are shown in Figure 6.A, and show no significant evidence of any non-parallel pre-treatment trends between states that opened recreational dispensaries and those that did not. A short time after recreational dispensaries open, there is a steady increase in the number of problem health days in treated states. This evidence supports the pre-treatment parallel trends assumption required for causal identification of the treatment effect found in Panel A of Table 7. It also suggests that there is some “take-up” time in the type of marijuana use that leads to health problems; or time for the effects of continual recreational marijuana use to unfold.

Figure 6.B depicts the event-study results for disrupted-activity days among respondents reporting physically unhealthy days but no mentally unhealthy days. There is no statistically significant evidence of non-parallel trends between treated and control states in the years before dispensaries open, with most point estimates near zero. After recreational dispensaries open, there is no significant increase in problem health days at any time. This pattern is consistent with the Panel B results in Table 7, showing that disrupted-activity days do not materially rise through physical health problems following dispensary openings.

Finally, Figure 6.C shows the event-study results for disrupted-activity days among respondents reporting mentally unhealthy days but no physically unhealthy days. There is no significant evidence of non-parallel trends at any point in the pre-period. After dispensaries open, there is a one-year lag before a sharp increase in disrupted-activity days in the second post-treatment year, after which the estimates remain elevated. The magnitude of the estimates is similar to those in panel A, further suggesting that the observed increase in disrupted-activity days is largely driven by mental health disruptions.

Summarizing, the results shown in Table 7 and Figure 6 provide evidence of significant increases in mental, but not physical, health problems that may interfere with work. This aligns with growing evidence from psychiatric and neuroscience research showing that frequent marijuana use may exacerbate some mental health problems, while also causing a range of negative cognitive impacts and persistent apathy (Hoch et al., 2025; Kroon et al., 2021; National Academies of Sciences, Engineering, and Medicine et al., 2017; Testai et al., 2022). Similarly, existing economics literature has shown negative impacts on mental health after RMLs, particularly for younger populations (Borbely et al., 2022). While research on older workers has shown improved physical capacity for work after RMLs (Abouk et al., 2021), my finding of a null effect on days of physical health disrupting usual activity indicates that the full working-age population does not generally receive the same benefits as older cohorts; this also aligns with work by Borbely et al. (2022). Thus, the impacts of recreational marijuana legalization may have made it more costly for some individuals to attend work, raising their propensity for absenteeism.

Table 7: ATT Estimates for effects of RML with recreational dispensaries on incidence of work-related health outcomes (BRFSS, 2002-2023)

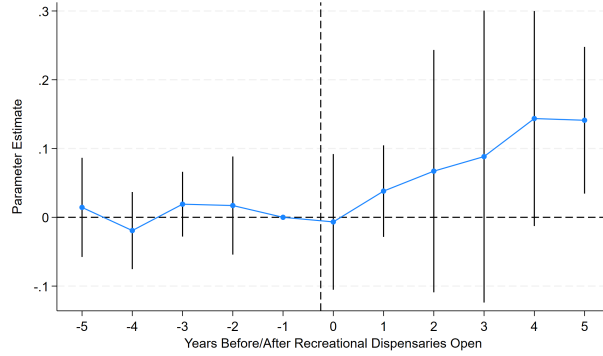
	(1)	(2)	(3)	(4)
	No Covariates	Add Demographics Controls	Add Macro & Social Policy Controls	Callaway Sant'Anna
Panel A: Days poor health prevented usual activities				
<i>RML with dispensaries</i>	0.0632** (0.0306)	0.0704** (0.0328)	0.0714** (0.0345)	0.0861** (0.0389)
Control Mean	1.2190	1.2190	1.2190	1.2190
State-year Observations	1,122	1,122	1,122	1,122
Panel B: Days disrupted exclusively by physical health problems				
<i>RML with dispensaries</i>	0.0067 (0.0128)	0.0073 (0.0131)	0.0069 (0.0136)	0.0092 (0.0116)
Control Mean	0.3990	0.3990	0.3990	0.3990
State-year Observations	1,122	1,122	1,122	1,122
Panel C: Days disrupted exclusively by mental health problems				
<i>RML with dispensaries</i>	0.0539*** (0.0185)	0.0580*** (0.0178)	0.0520*** (0.0178)	0.0625*** (0.0186)
Control Mean	0.2739	0.2739	0.2739	0.2739
State-year Observations	1,122	1,122	1,122	1,122
Demographic Controls	No	Yes	Yes	No
Social Welfare Policies	No	No	Yes	No
Macroeconomic Controls	No	No	Yes	No
Model	S&A	S&A	S&A	CS

* $p < 0.10$, ** $p < .05$, *** $p < .01$.

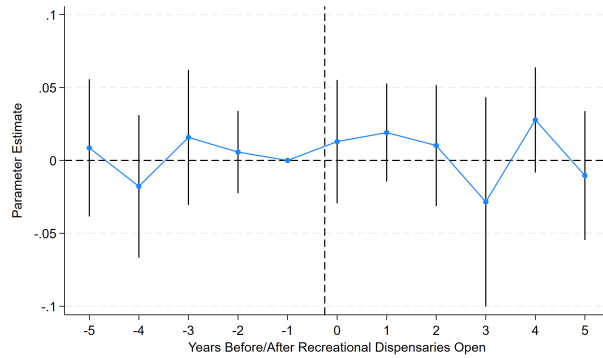
Notes: Each column represents a separate regression. Results are derived using Sun and Abraham's (2021) dynamic treatment effect estimator. Data from the 2002-2023 BRFSS and include all observations aged 18-64 who report being employed. All regressions include state and year fixed effects, and are weighted by state population using SEER data. Standard errors, clustered at the state level, are in parentheses.

Figure 6: Sun & Abraham Event Studies, Effects of RML with recreational dispensaries on work-related health outcomes (BRFSS, 2002-2023)

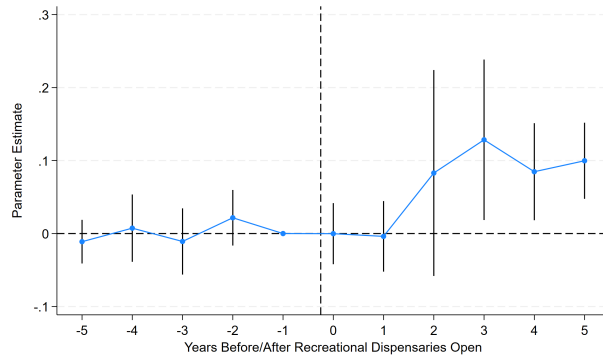
(a): Days poor health prevented usual activities



(b): Days disrupted exclusively by physical health problems



(c): Days disrupted exclusively by mental health problems



Notes: The graphs contains coefficient estimates with 95% confidence intervals for years before and after the change in RML, derived using Sun and Abraham (2021). The models control for all covariates, state and time fixed effects, and are weighted by state population. Standard errors are clustered at the state level. Data are from the 2002-2023 BRFSS and include all employed observations aged 18-64.

7 Discussion

This study provides new causal evidence on the effects of recreational marijuana policy implementation on workplace absenteeism. First, using NSDUH data, I show that the rollout of recreational dispensaries significantly increases marijuana use beyond what is observed under medical marijuana laws and decriminalization. Specifically, I find that recreational dispensary openings lead to an approximate 50% increase in the prevalence of self-reported marijuana consumption over the control group mean. These results are similar to recent literature (Dave et al., 2023; Sabia et al., 2024).

Next, the main analysis shows that state RML adoption with operational dispensaries significantly increases rates of workplace absence. Among the employed population aged 18-64, I find an overall increase of 0.45 percentage points in the self-reported incidence of prior-week absence from work, a 14% increase over the control group mean. An extensive set of sensitivity analyses indicate that these results are not driven by Covid-19 or childcare needs. Dynamic event study estimates, which adjust for staggered policy implementation, lend support for a causal interpretation of these results. As illustrated in Figure 3, the absenteeism increase is only evident when treatment is defined by the onset of legal retail sales, whereas estimates based on law passage are small and statistically indistinguishable from zero.

Supplemental analyses using BRFSS data provide evidence consistent with a potential health-based mechanism, showing that dispensary openings modestly increase mental health problems that disrupt usual activity. This mechanism may explain the increase in sickness-related absences, which make up about a third of the overall increase in absenteeism. The increase in unplanned vacation or personal time may reflect other behavioral responses to policy implementation, including changes in work-leisure choices (Hoch et al., 2025). Further research is needed to examine the mechanisms contributing to the finding of increased work absence.

These findings align with emerging patterns in the literature showing differential effects of marijuana legalization for different groups. The effect of increased workplace absence may be concentrated among a subset of recreational consumers rather than those who use it to successfully treat health problems. This suggests that a subset of the workforce population between ages 18-64 may have differentiated risk factors for negative mental health or motivational responses to recreational use, translating to increased unplanned absenteeism from work. This is in comparison to those who use marijuana to treat health issues, particularly older populations, who were observed to benefit from the positive effects of medical and recreational marijuana legalization in existing research on mental health, health, and employment outcomes (Abouk et al., 2021; Borbely et al., 2022; Drake et al., 2025; Ullman, 2017).

Collectively, this evidence suggests that recreational marijuana policy implementation may contribute to higher rates of workplace absenteeism. As such, they represent one potential consequence of expanding recreational marijuana access through state policy. While this work does expose a negative outcome on the intensive margin, further study may be necessary to interpret the long-term effects of increased absence on overall employee welfare.

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A Appendix Tables

Table A1: State Marijuana Policy Timeline

State	MML Effective Date	RML Effective Date	Recreational Sales Allowed
Alaska	03/04/1999	02/24/2015	10/29/2016
Arkansas	11/09/2016	-	-
Arizona	04/14/2011	11/30/2020	01/22/2021
California	11/06/1996	11/09/2016	01/01/2018
Colorado	06/01/2001	12/10/2012	01/01/2014
Connecticut	08/20/2014	07/01/2021	01/10/2023
Delaware	06/26/2015	-	-
District of Columbia	07/30/2013	02/26/2015	02/26/2015
Florida	07/26/2016	-	-
Hawaii	12/28/2000	-	-
Illinois	11/09/2015	01/01/2020	01/01/2020
Louisiana	08/06/2019	-	-
Maine	12/22/1999	01/31/2017	10/09/2020
Maryland	12/02/2017	-	-
Massachusetts	01/01/2013	12/15/2016	11/20/2018
Michigan	12/04/2008	12/06/2018	12/01/2019
Minnesota	07/01/2015	-	-
Mississippi	02/02/2022	-	-
Missouri	10/17/2020	12/08/2022	02/03/2023
Montana	11/02/2004	01/01/2021	01/01/2022
Nevada	10/01/2001	01/01/2017	07/01/2017
New Hampshire	05/01/2016	-	-
New Jersey	12/06/2012	02/22/2021	04/21/2022
New Mexico	07/01/2007	06/29/2021	04/01/2022
New York	01/08/2016	03/31/2021	12/29/2022
North Dakota	03/01/2019	-	-
Ohio	01/16/2019	-	-
Oklahoma	07/26/2018	-	-
Oregon	12/03/1998	07/01/2015	10/01/2015
Pennsylvania	01/17/2018	-	-
Rhode Island	01/03/2006	5/25/2022	12/01/2022
South Dakota	07/01/2021	-	-
Utah	03/02/2020	-	-
Vermont	07/01/2004	07/01/2018	10/01/2022
Virginia	10/17/2020	07/01/2021	-
Washington	11/03/1998	12/06/2012	07/08/2014
West Virginia	08/22/2017	-	-

Notes: all original policy dates are drawn directly from (Anderson & Rees, 2023). Updated dates, in bold print, are drawn from the author's research into legislative changes.

Table A2: Types of Non-layoff Absences
CPS ASEC (2002-2023), Ages 18-64

	Count	Share	Included in Main Analysis
Reason for Workplace Absence			
<i>Vacation or Personal Days</i>	29,604	41.4%	Yes
<i>Own Illness, Injury, or Medical Problems</i>	16,428	22.3%	Yes
<i>Other Personal Obligation</i>	4,447	6.22%	Yes
<i>Weather Affected Job</i>	2,944	4.12%	Yes
<i>School or Training</i>	1,747	2.44%	Yes
<i>Other</i>	11,388	15.92%	Yes
<i>Maternity or Paternity Leave</i> ¹	4,454	6.23%	No
<i>Child Care Problems</i> ²	467	0.65%	No
<i>Labor Dispute</i>	94	0.14%	No
<i>Civic or Military Duty</i>	130	0.18%	No
Total Count	71,523		

¹ I exclude observations reporting maternity or paternity leave as the reason for work absence from the main analysis. These types of absences are typically pre-planned and documented, and are thus unlikely to be influenced by marijuana liberalization. Additionally, access to parental leave and childcare may vary systematically across states, particularly between those that adopted RMLs and those that did not.

² Absences due to childcare problems are also excluded to mitigate potential confounding from cross-state variation in childcare availability and parental leave policies between treatment and control units.

Table A3: Conventional TWFE Estimates for effects of RML with recreational dispensaries on main outcomes

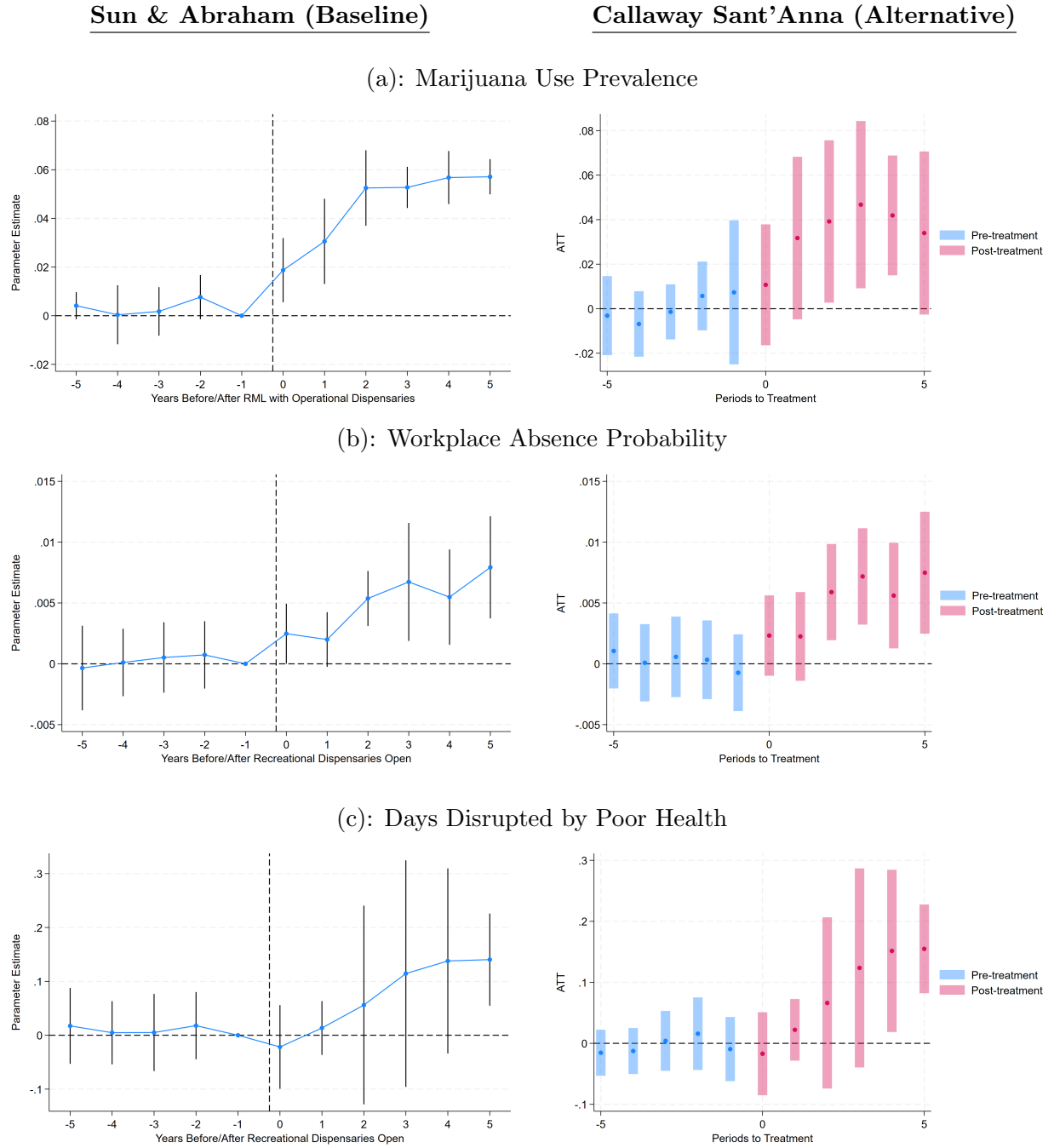
	(1)	(2)
	Sun & Abraham	Two Way Fixed Effects
	(Baseline)	(Alternative)
Panel A: Any marijuana use		
<i>RML with dispensaries</i>	0.0373***	0.0389***
	(0.0046)	(0.0077)
Control Mean	0.0713	0.0736
State-year Observations	867	867
Panel B: Absent from work		
<i>RML with dispensaries</i>	0.0045***	0.0029***
	(0.0012)	(0.0009)
Control Mean	0.0310	0.0292
Individual Observations	1,774,284	1,774,284
Panel C: Days poor health prevented usual activities		
<i>RML with dispensaries</i>	0.0714**	0.0624**
	(0.0345)	(0.0264)
Control Mean	1.2190	1.2344
State-year Observations	1,122	1,122
Demographic Controls	Yes	Yes
Social Welfare Policies	Yes	Yes
Macroeconomic Controls	Yes	Yes
Model	S&A	TWFE

* $p < 0.10$, ** $p < .05$, *** $p < .01$.

Notes: Each column represents a separate regression. Results in column (1) are derived using the interaction weighted estimator proposed by Sun and Abraham (2021), while those in column (2) are derived from a conventional two-way fixed effects (TWFE) model estimated by OLS. Data are from the 2002-2023 BRFSS and include all observations aged 18-64 who report being employed. Both regressions include state and year fixed effects along with the full set of controls used in the main specification. Standard errors, clustered at the state level, are in parentheses.

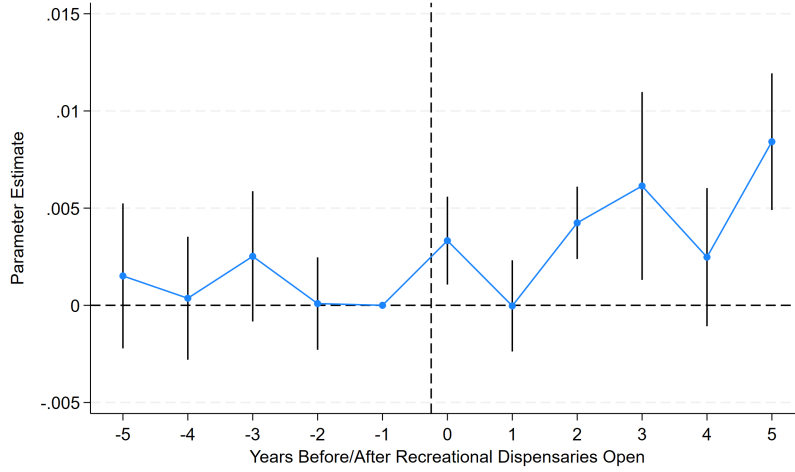
B Appendix Figures

Figure B1: Callaway Sant'Anna Event Studies



Notes: The graphs contain coefficient estimates with 95% confidence intervals for years before and after recreational marijuana sales begin. The models control for all covariates, state and time fixed effects. Standard errors are clustered at the state level. The analyses on the left use Sun and Abraham (2021). Those on the right use Callaway and Sant'Anna (2021). The models are otherwise identical.

Figure B2: Robustness to Cross-border Spillovers, Workplace Absence



Notes: To test sensitivity to possible geographic spillovers, I re-estimate the main event-study excluding from the control group any states that border a recreational dispensary or recreational law state. This leaves only eight geographically isolated control states: MN, HI, AL, FL, GA, LA, MS, and SC. The resulting dynamic estimates are qualitatively similar to the baseline specification that retains bordering controls, suggesting limited contamination from cross-border spillovers. Points report Sun and Abraham (2021) estimates with 95% confidence intervals. The regression includes state and time fixed effects, and is weighted by CPS ASEC sampling weights. Standard errors are clustered at the state level.

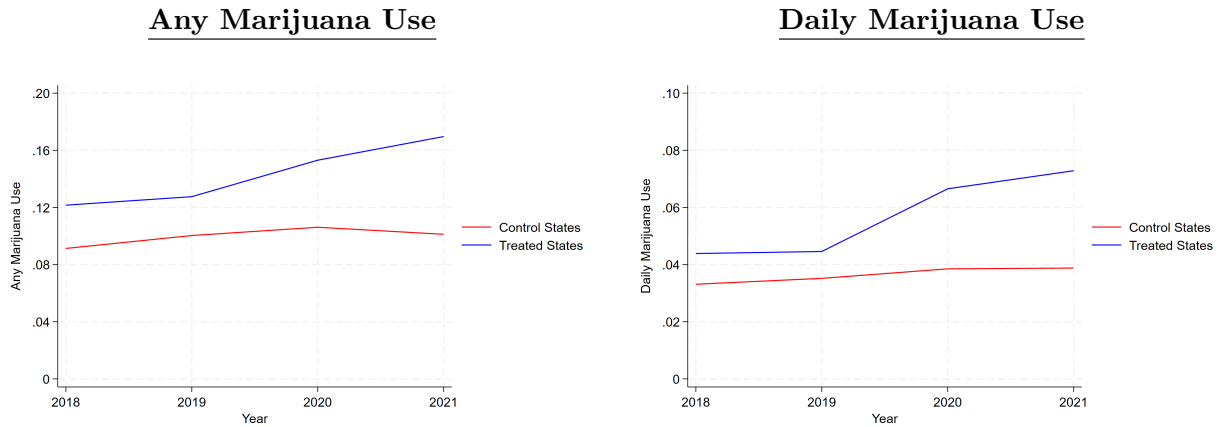
ATT Estimates for cross-border spillover test (CPS ASEC, 2002-2023)

	(1) Baseline Estimate	(2) Excluding Border States
Absent from Work		
<i>RML with dispensaries</i>	0.0045*** (0.0012)	0.0037*** (0.0011)
Control Mean	0.0310	0.0267
Individual Observations	1,774,284	1,047,770
Model	S&A	S&A

* $p < 0.10$, ** $p < .05$, *** $p < .01$.

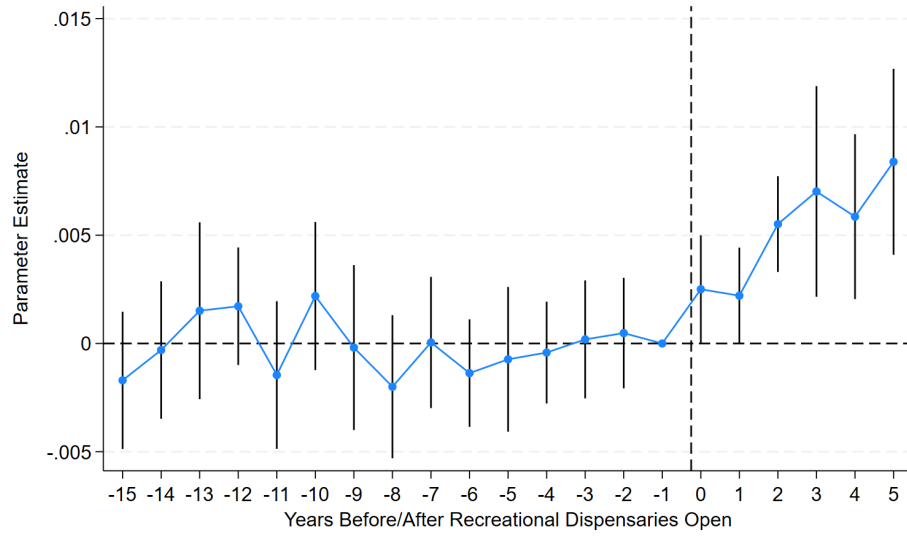
Notes: Each column represents a separate regression. Results are derived using Sun and Abraham's (2021) dynamic treatment effect estimator to account for timing-induced bias. Data are from the 2002-2023 CPS ASEC. Both models include state and year fixed effects, along with the full set of covariates as per column (3) of Table 2. Standard errors, clustered at the state level, are in parentheses.

Figure B3: Trends in Any and Daily Marijuana Use (BRFSS, 2018-2021)



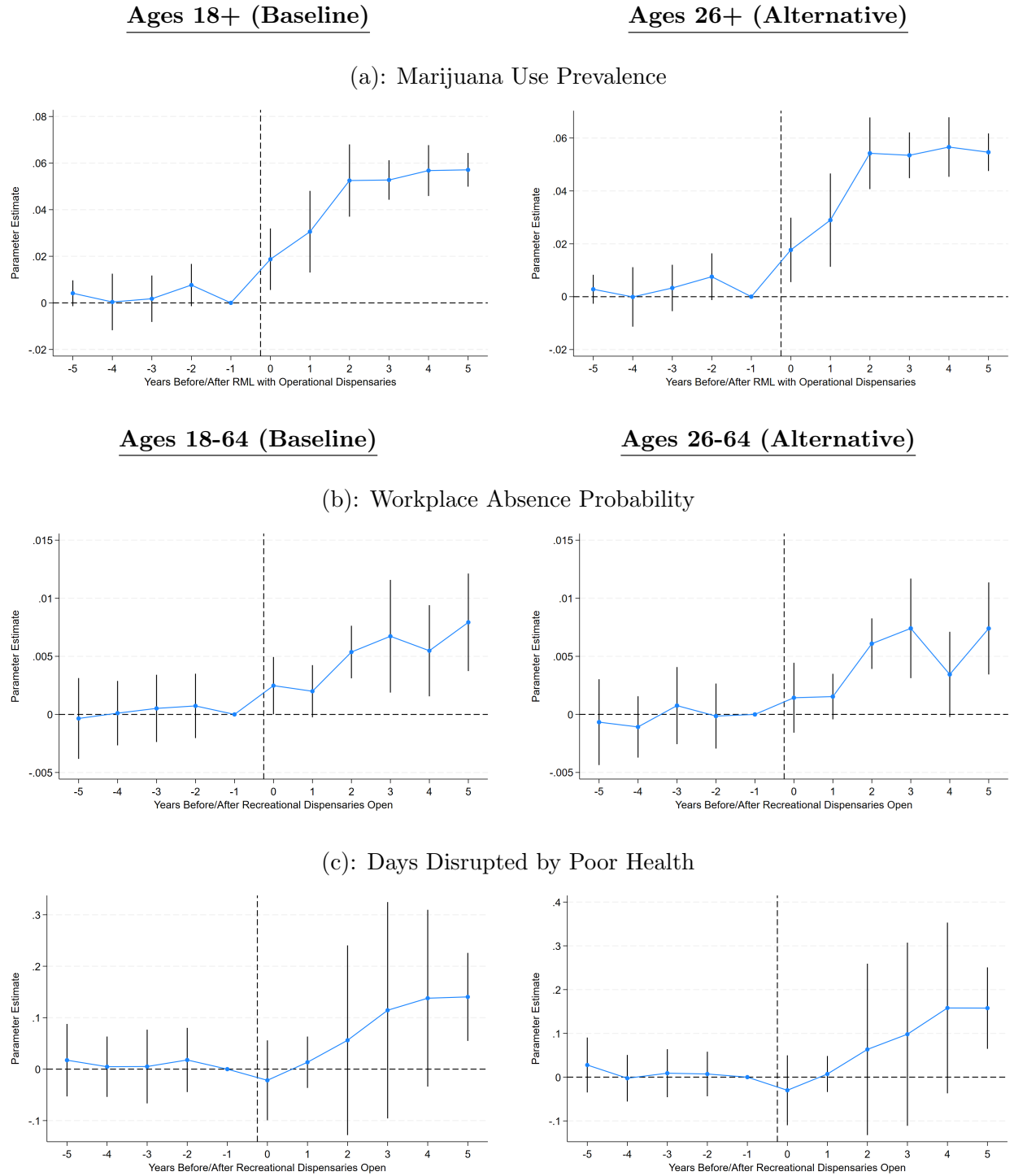
Notes: The figure plots annual weighted means of marijuana use from the BRFSS optional marijuana module for the subset of state-years in which the questions were asked. The figure on the left shows the prevalence of any marijuana use in the past 30 days, and the figure to the right shows the prevalence of daily use among past-month users. Estimates are only available for a limited subset of state-years that fielded the module. These trends are presented for descriptive context and are not interpreted as causal effects of legalization.

Figure B4: Extended Event Study, Workplace Absence



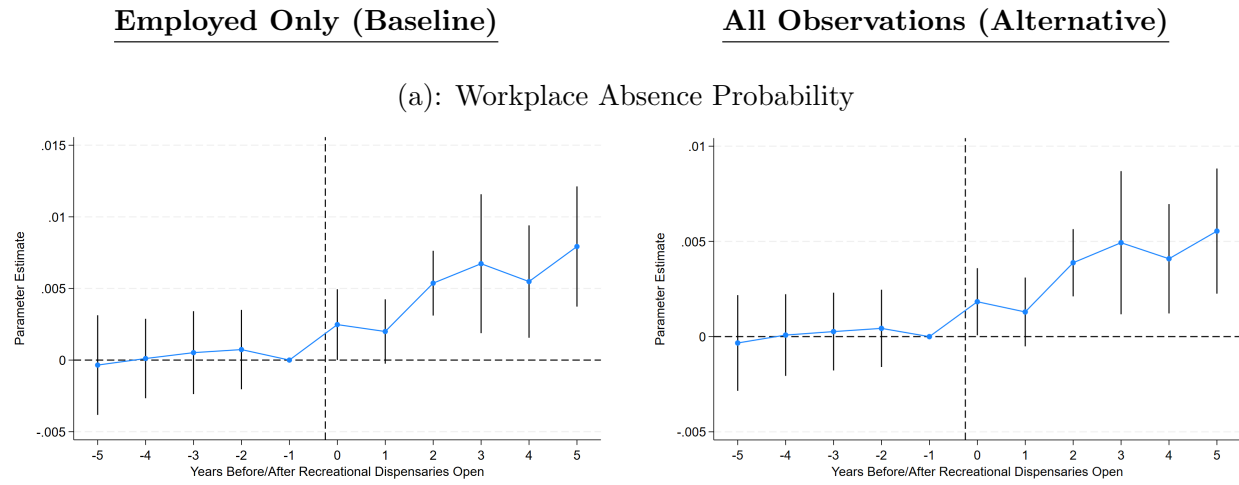
Notes: As an additional diagnostic for the parallel trends assumption, I re-estimate the main event-study specification extending the pre-treatment window to include an additional 10 years of leads. The resulting estimates show no statistically significant differences between treated and control states in the extended pre-period. Points report Sun and Abraham (2021) estimates with 95% confidence intervals. The regression includes all covariates, as well as state and year fixed effects, and is weighted by CPS ASEC sampling weights. Standard errors are clustered at the state level.

Figure B5: Event Studies, Effects of recreational marijuana dispensaries on main outcomes (Alternative Quasi-experiment)



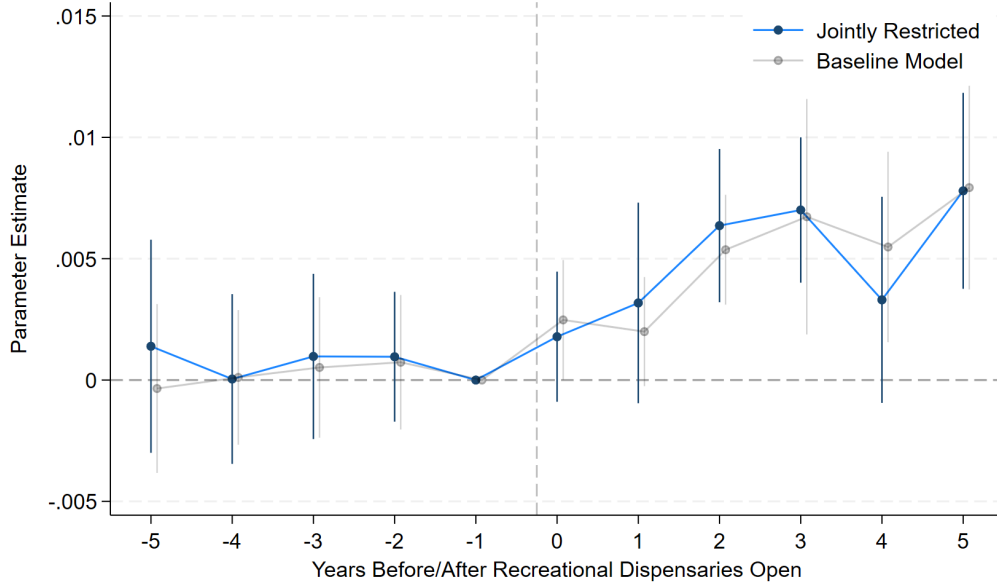
Notes: The graphs contain coefficient estimates with 95% confidence intervals for years before and after the change in RML, derived using Sun and Abraham's (2021) dynamic treatment effect estimator to account for timing-induced bias. The regressions control for all covariates, state and time fixed effects. The analyses on the left use the original aged 18-64 sample. Those on the right restrict the sample to ages 26-64. The models are otherwise identical.

Figure B6: Event Study, Effect of RML with recreational dispensaries on absence probability for the employed, plus the unemployed and those not in the labor force (Alternative Quasi-experiment)



Notes: The graphs contain coefficient estimates with 95% confidence intervals for years before and after the change in RML, derived using Sun and Abraham's (2021) dynamic treatment effect estimator to account for timing-induced bias. The regressions control for all covariates, state and time fixed effects. The analyses on the left use the original employed-only sample. Those on the right expand the sample to also include the unemployed and those not in the labor force. The models are otherwise identical.

Figure B7: Robustness to Joint Sample Restrictions, Workplace Absence



Notes: The graph contains coefficient estimates with 95% confidence intervals for years before and after the onset of legal recreational sales, derived using Sun and Abraham’s (2021) dynamic treatment effect estimator. The highlighted specification jointly imposes three sample restrictions: event time balance¹, the exclusion of border states², and the exclusion of calendar year 2020³ to simultaneously mitigate three major threats to identification. The main specification (Figure 2) is presented in translucent gray for comparison. The dependent variable is an indicator for prior-week workplace absence. The models include all covariates, state and time fixed effects as per column (3) of Table 3. Standard errors are clustered at the state level. Data are from the 2002-2023 CPS ASEC. Despite the substantial sample restrictions, the estimated dynamic effects closely track the baseline results. This jointly restricted specification is presented as a conservative robustness check.

¹ To address concerns about event-time compositional change, I construct a restricted panel of treated states that is balanced in event time from -5 to $+5$ relative to the year of recreational dispensaries opening. This includes the full subset of states that contribute identifying variation to all of these relative years: AK, CA, CO, DC, NV, OR, and WA.

² To address concerns about geographic spillovers, I exclude from the control group all states that border a legal recreational state at any point during the sample period. This restriction yields a never-treated comparison group that is less likely to be contaminated by cross-border access to recreational cannabis markets: AL, FL, GA, HI, LA, MN, MS, and SC.

³ To address concerns about COVID-era confounding, I also exclude calendar year 2020 from this analysis.

C Additional Testing for Parallel Trends

In this section, I evaluate the sensitivity of my event study results to potential violations of the parallel trends assumption. I apply the framework proposed by Rambachan and Roth (2023). Below, I briefly summarize their approach and explain how I apply it to this setting. The exposition in this appendix draws on Appendix C of my related working paper (M. Brown, 2025); only the outcomes and resulting robustness thresholds differ in this paper.

Rambachan and Roth (2023) define δ as the counterfactual difference in trends between treated and control groups in the absence of treatment. They then consider a researcher-specified set Δ that bounds deviations from parallel trends. For any candidate Δ , they derive robust confidence intervals that account for these possible violations. They propose using two benchmark specifications for Δ .

The first approach, *Relative Magnitude Bounds*, assumes that post-treatment deviations from parallel trends are not substantially greater than those observed in the pre-treatment period. Formally,

$$\Delta_{\text{RM}}(\bar{M}) = \left\{ \delta : \forall t \geq 0, |\delta_{t+1} - \delta_t| \leq \bar{M} \cdot \max_{s < 0} |\delta_{s+1} - \delta_s| \right\}, \quad (2)$$

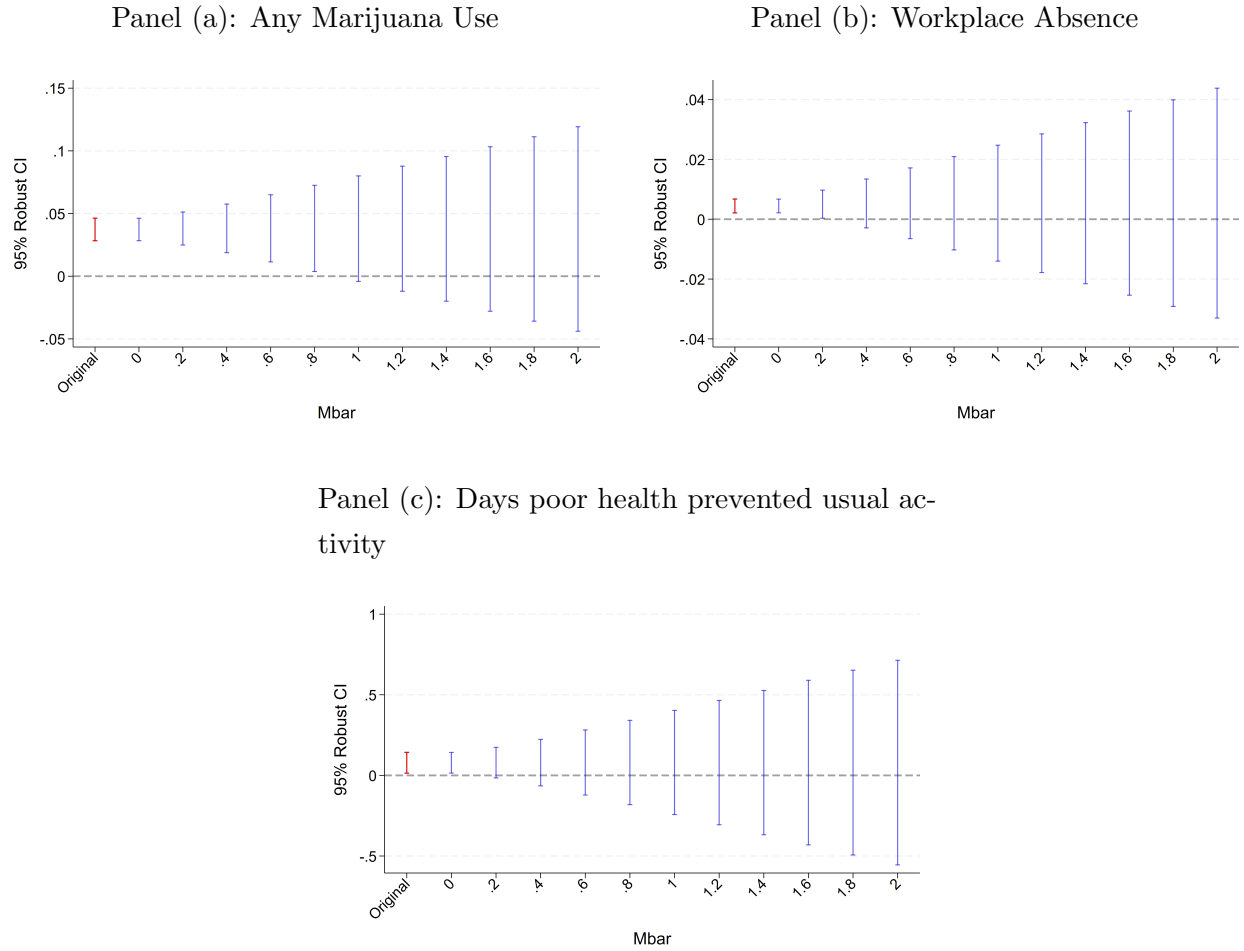
where $\bar{M}$ scales the allowable post-treatment deviation relative to the maximum pre-treatment change. The second specification, *Smoothness Restrictions*, requires differential trends to evolve smoothly over time by bounding second differences in δ :

$$\Delta_{\text{SD}}(M) = \{ \delta : |(\delta_{t+1} - \delta_t) - (\delta_t - \delta_{t-1})| \leq M, \forall t \}, \quad (3)$$

where $M = 0$ implies a linear trend in the differences between treated and control groups. I use both sets of restrictions to gauge the sensitivity of the treatment effects estimated from equation (1). As emphasized by Rambachan and Roth (2023), I focus on the robustness of the average post-treatment effect over the full event-time window used in the main analysis, rather than on the first post-treatment coefficient alone. Accordingly, I report robust confidence intervals for this average effect. For the relative magnitude bound, I set $\bar{M} = 2$, permitting post-treatment deviations up to twice the largest pre-treatment change. For the smoothness restriction, I adopt the default scaling in the `honestDiD` Stata package, which normalizes M to the scale of the outcome.

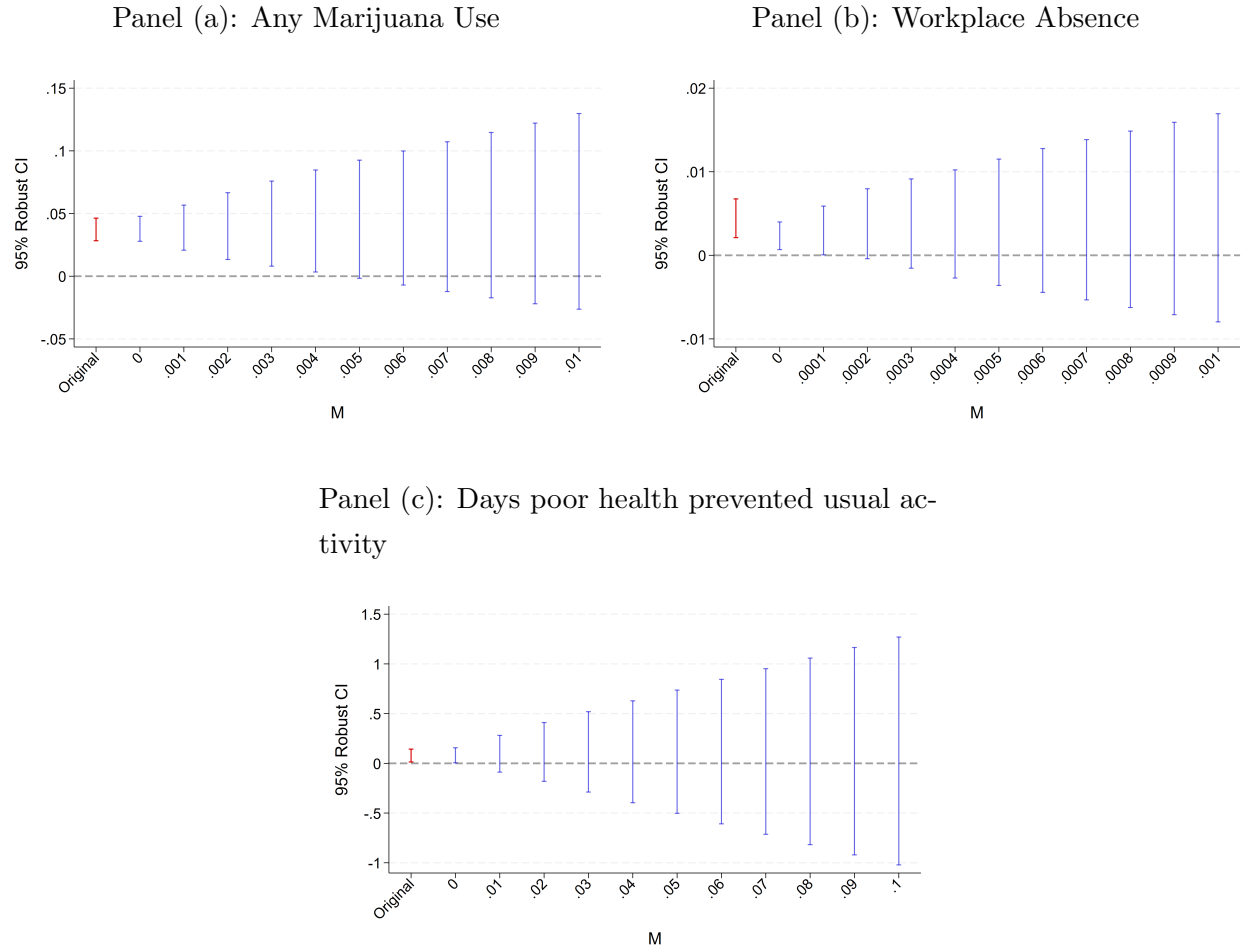
Appendix Figures C1 and C2 show 95% confidence intervals from each test. Figure C1 displays estimates under the relative magnitude restriction. The estimates remain robust to post-treatment violations of the parallel trends assumption up to 80% of the maximum pre-period deviation for marijuana use, up to 20% for workplace absence, and up to 10% for disrupted-activity days. Figure C2 reports results under the smoothness restriction. For all outcomes, the estimated effects remain robust under the $M = 0$ restriction.

Figure C1: Sensitivity Analysis for Non-parallel Trends (Relative Magnitude Restrictions)



Notes: These graphs display 95% robust confidence intervals for the treatment effects of recreational marijuana dispensary openings on the listed outcomes, estimated using the “relative magnitude bounds” sensitivity analysis framework developed by (Rambachan & Roth, 2023). Panels a through c report results for the average treatment effect on the treated (ATT), calculated as a weighted average of cohort-specific post-period treatment effects (CATTs) across relative event times. Baseline ATT estimates under parallel trends are shown in red.

Figure C2: Sensitivity Analysis for Non-parallel Trends (Smoothness Restrictions)



Notes: These graphs display 95% robust confidence intervals for the treatment effects of recreational marijuana dispensary openings on the listed outcomes, estimated using the "smoothness restriction" sensitivity analysis framework developed by (Rambachan & Roth, 2023). Panels a through c report results for the average treatment effect on the treated (ATT), calculated as a weighted average of cohort-specific post-treatment treatment effects (CATTs) across relative event times. Baseline ATT estimates under parallel trends are shown in red.